When RPKI Meets Reality:

Understanding the Causes and Risks of RPKI-Invalid

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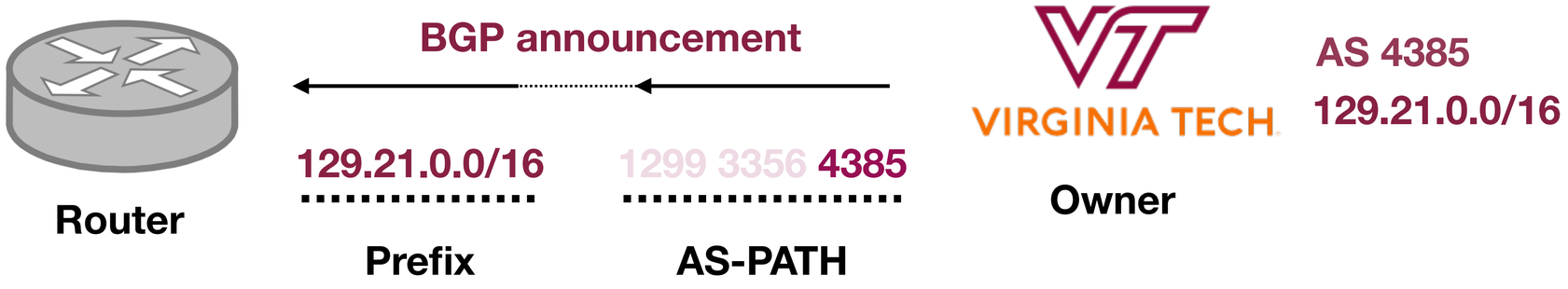
[1]Virginia Tech, [2]CableLabs

Border Gateway Protocol (BGP)

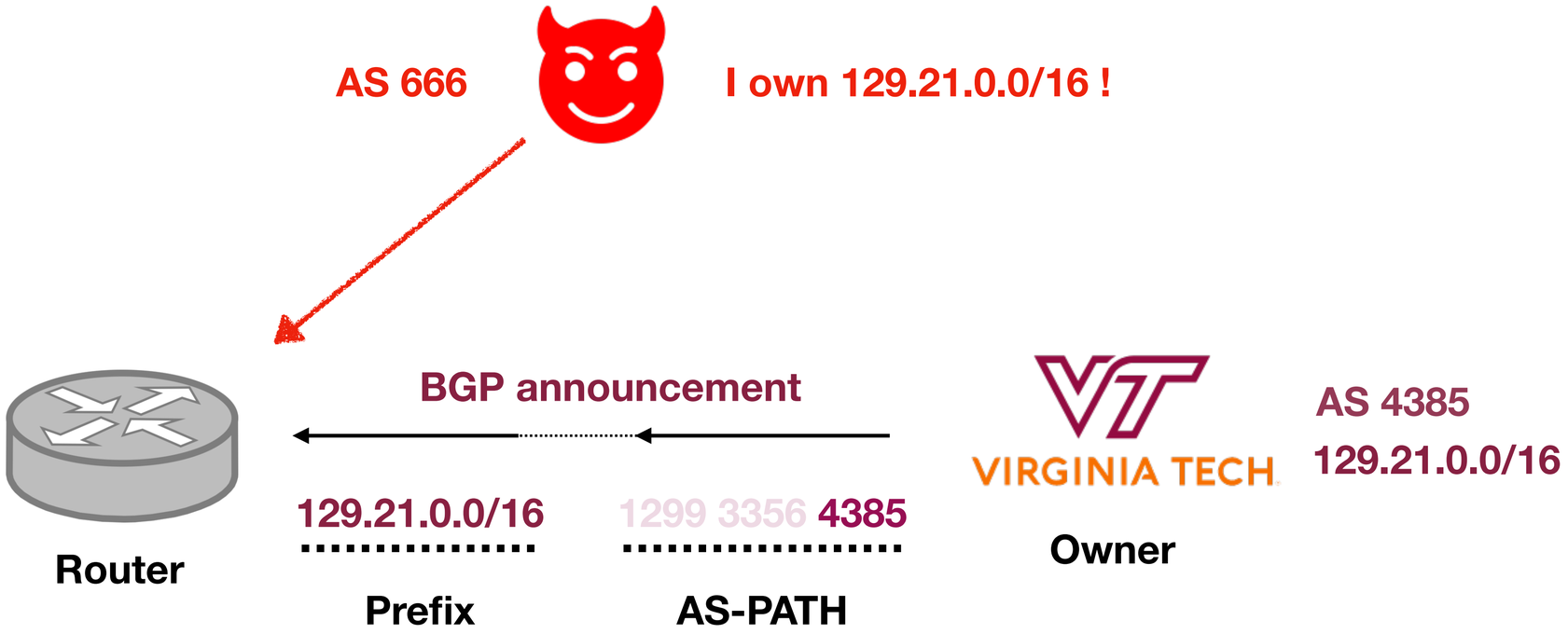
- Each network resource owner (e.g., VT) announces its IP prefixes to the rest of routers, so that they can learn the path towards VT.

- However, BGP does not have builtin security mechanism



Border Gateway Protocol (BGP)

- An adversary can announce prefixes that not belong to it, thus hijack the resource



BGP Hijacks

THE POWER OF FALSE ADVERTISING —

How an Indonesian ISP took down the mighty Google for 30 minutes

Internet's web of trust let a company you never heard of block your Gmail.

SEAN GALLAGHER - 11/6/2012, 11:07 AM

Google's services went offline for many users for nearly a half-hour on the evening of November 5, thanks to an erroneous routing message broadcast by Moratel, an Indonesian telecommunications company. The outage might have lasted even longer if it hadn't been spotted by a network engineer at CloudFlare who had a friend in a position to fix the problem.

The root cause of the outage was a configuration change to routers by Moratel, apparently intended to block access to Google's services from within Indonesia. The changes used the Border Gateway Protocol to "advertise" fake routes to Google servers, shunting traffic off to nowhere. But because of a misconfiguration, the BGP advertisements "leaked" through a peering connection in Singapore and spread to the wider Internet through Moratel's connection to the network of Hong Kong-based backbone provider PCCW. Google was interrupted in a similar way in 2008, when Pakistan Telecom moved to block access to YouTube in Pakistan because of an order from the Pakistani government.

Tom Paseka, a networking engineer at the content distribution network and Web security provider Cloudflare, spotted the source of the outage. "When I figured out the problem," Paseka wrote in CloudFlare's blog this morning, "I contacted a colleague at Moratel to let him know what was going on. He was able to fix the problem at around 2:50 UTC / 6:50pm PST. Around 3 minutes later, routing returned to normal and Google's services came back online."

TECH | CYBERSECURITY | CRYPTOCURRENCY

Hackers emptied Ethereum wallets by breaking the basic infrastructure of the internet

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By Russell Brandom | @russellbrandom | Apr 24, 2018, 1:40pm EDT

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The Solution: RPKI

- Resource Public Key Infrastructure (RPKI) is proposed to secure Internet's routing and prevent hijacks
- The deployment of RPKI starts from 2008 and received more

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- Resource Public Key Infrastructure (RPKI) is proposed to secure Internet's routing and prevent hijacks
- The deployment of RPKI starts from 2008
- RPKI contains two parts:
  - Route Origin Authorization (ROA)
  - Route Origin Validation (ROV)

RPKI Structure: ROA

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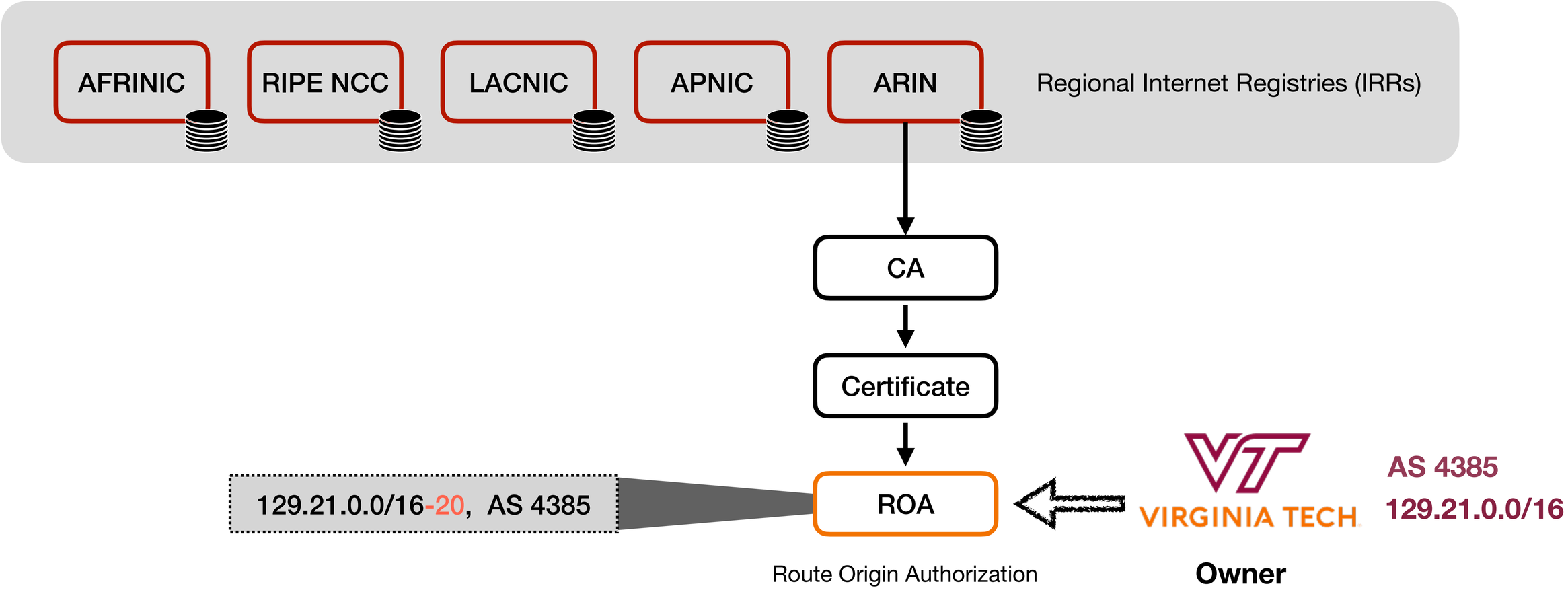
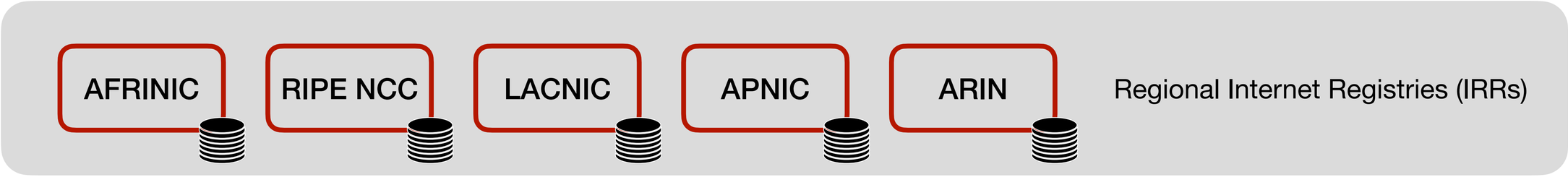
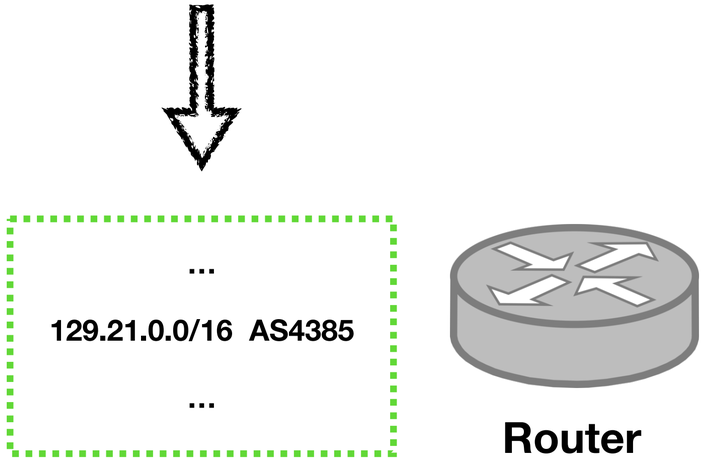


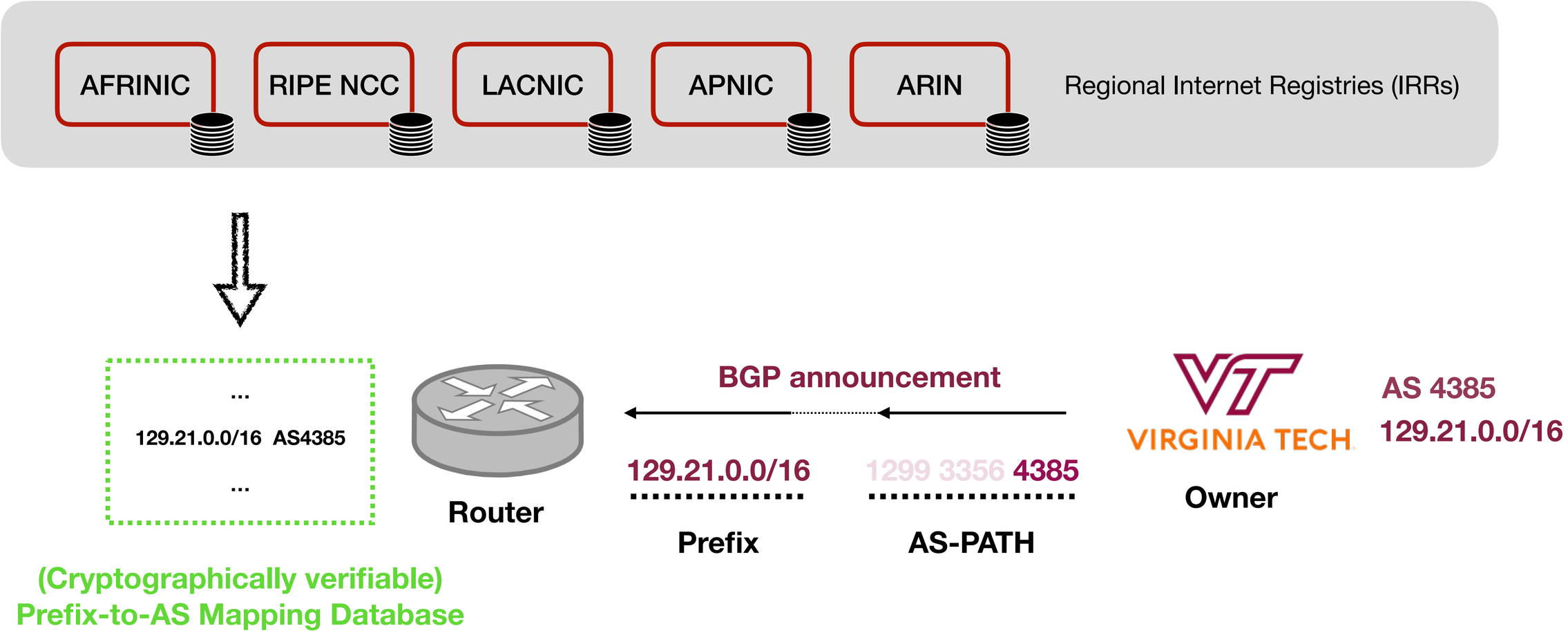
RPKI Structure: Route Origin Validation



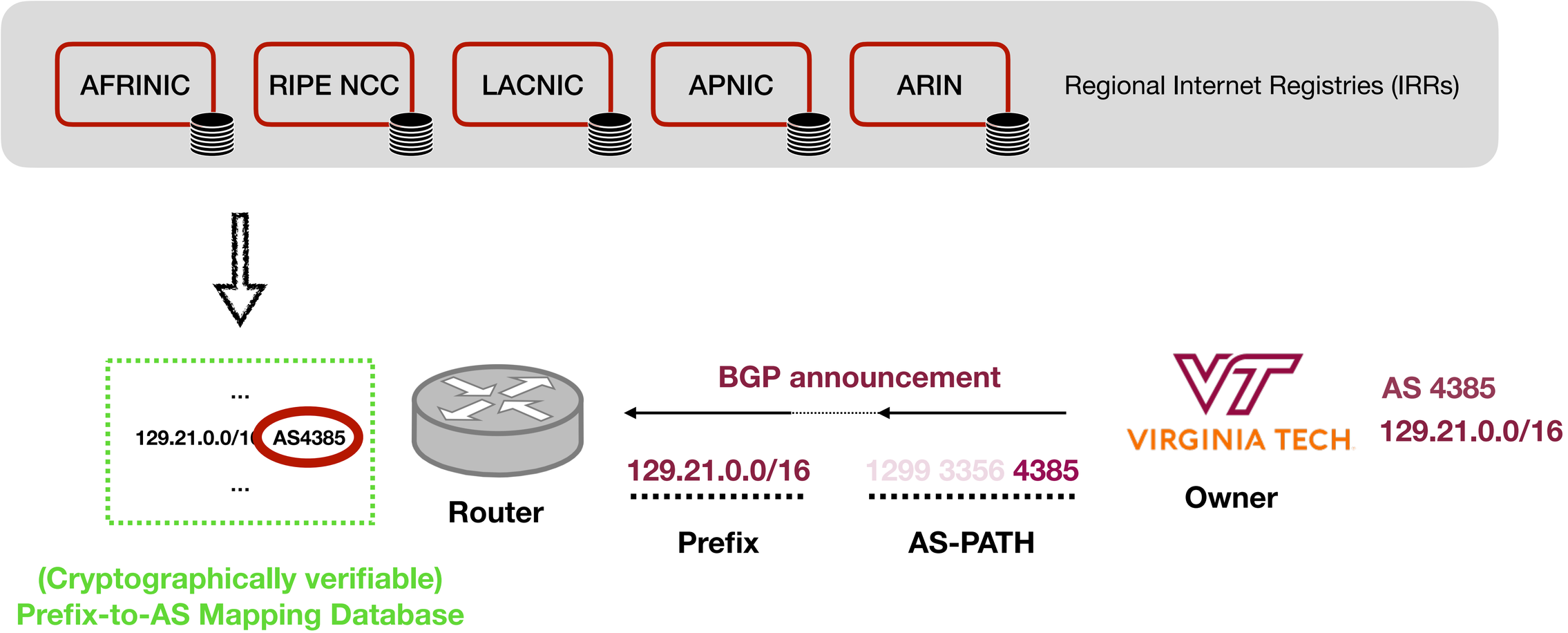


(Cryptographically verifiable)
Prefix-to-AS Mapping Database

RPKI Structure: Route Origin Validation



RPKI Structure: ROV



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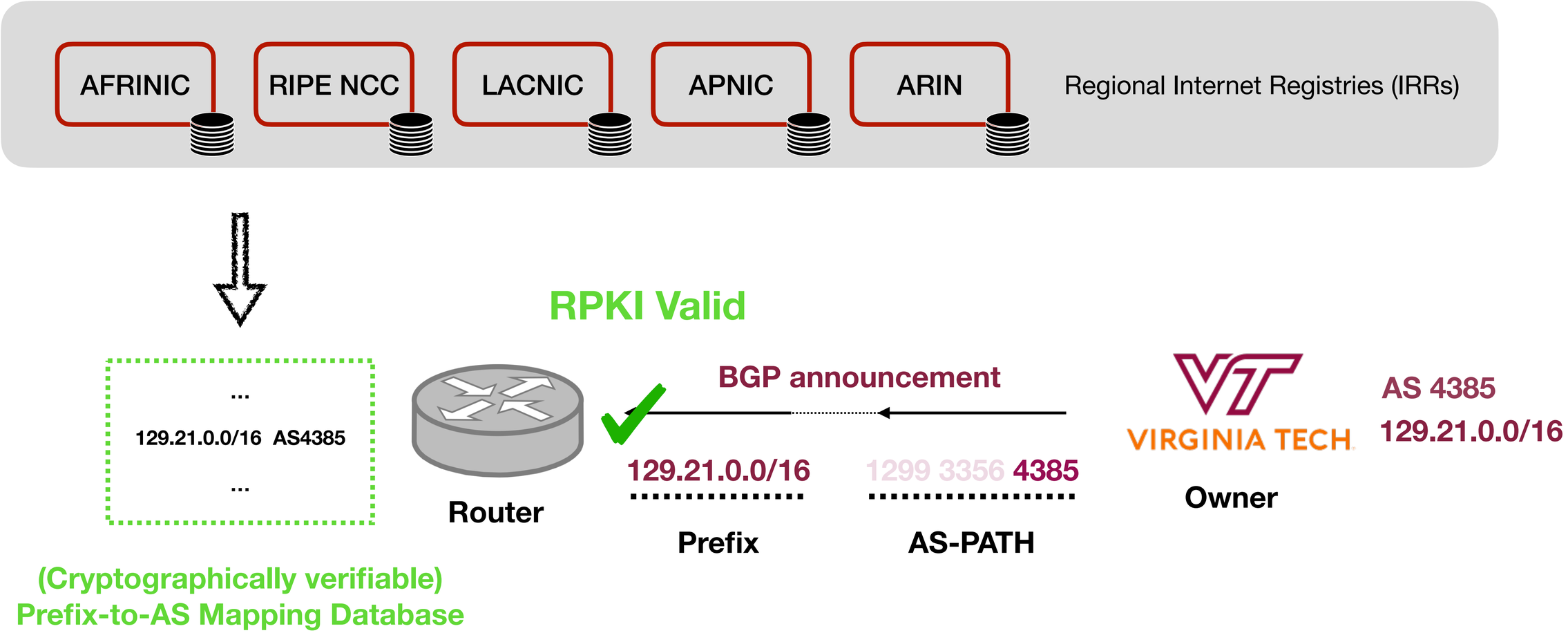

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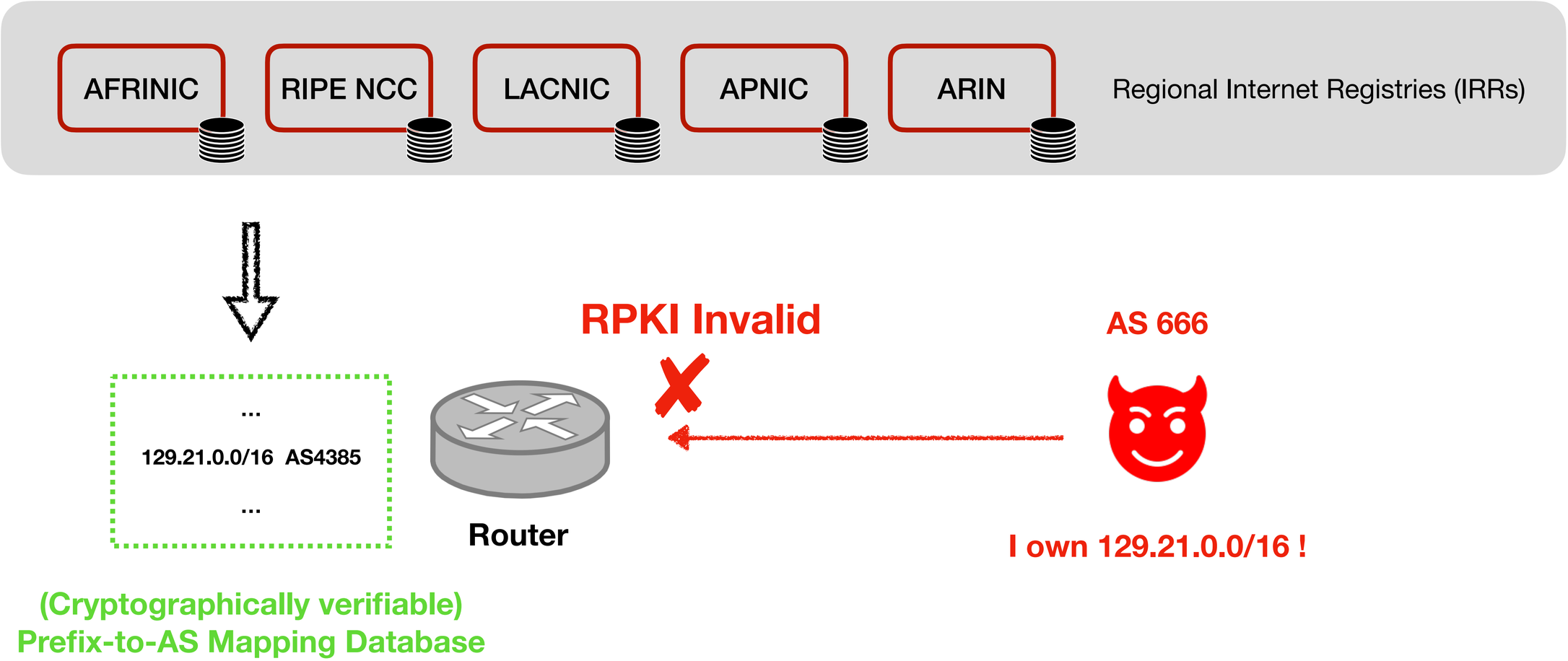
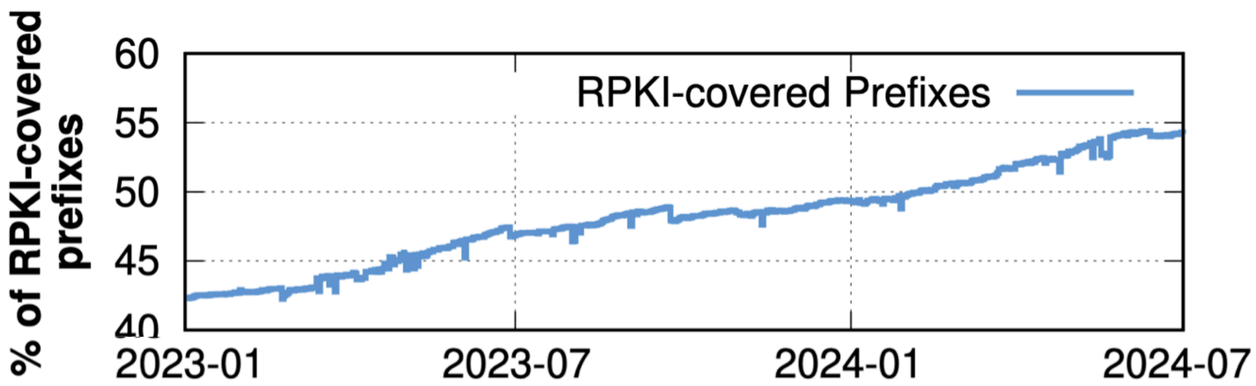


The Deployment of ROA

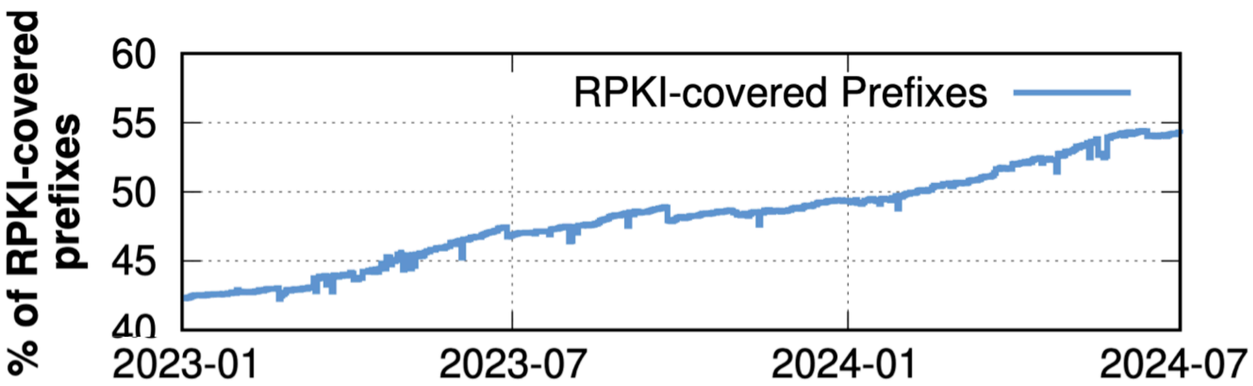
- We analyze BGP routing tables from all collectors of RouteViews and RIPE RIS during a 18-month period



The Deployment of ROA

- We analyze BGP routing tables from all collectors of RouteViews and RIPE RIS during a 18-month period

- The coverage of ROAs are growing, 54% of the IPv4 spaces are covered by ROAs. More networks are deploying ROA, but ….



RPKI-Invalid routes never go away

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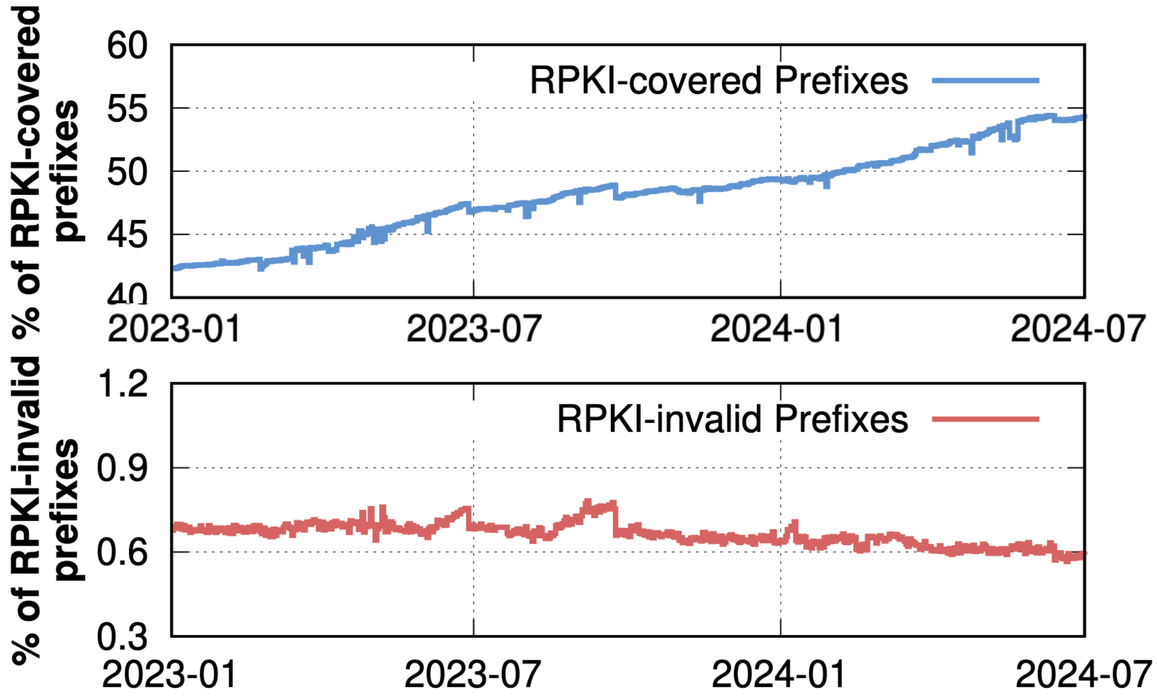

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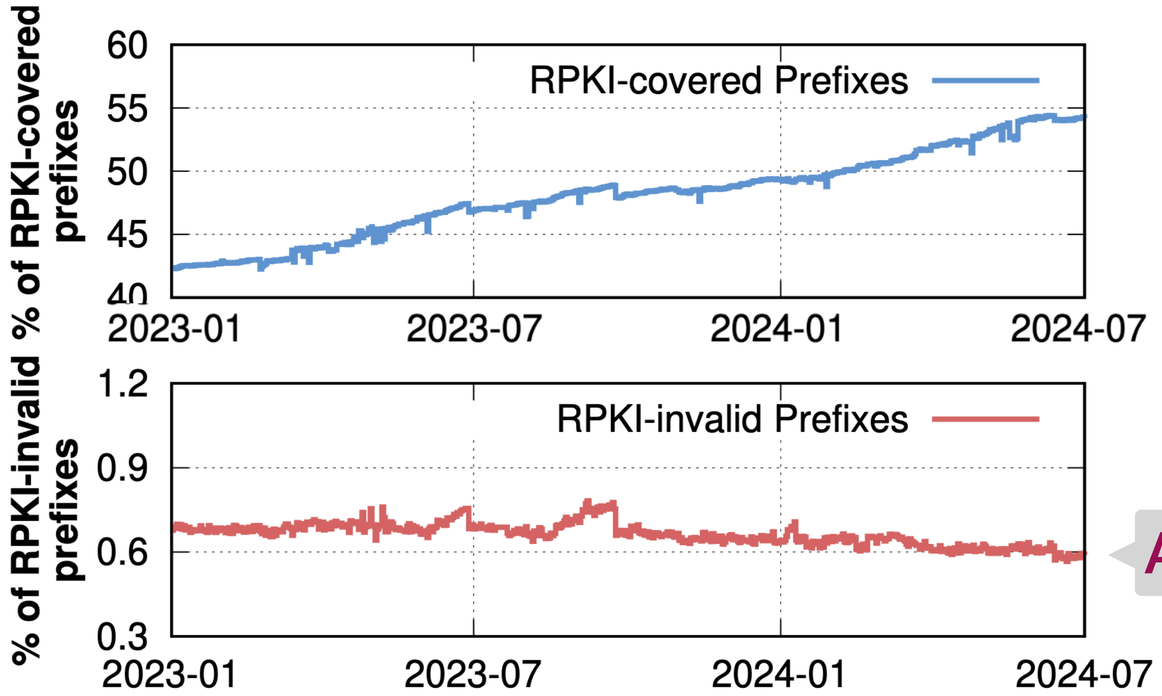
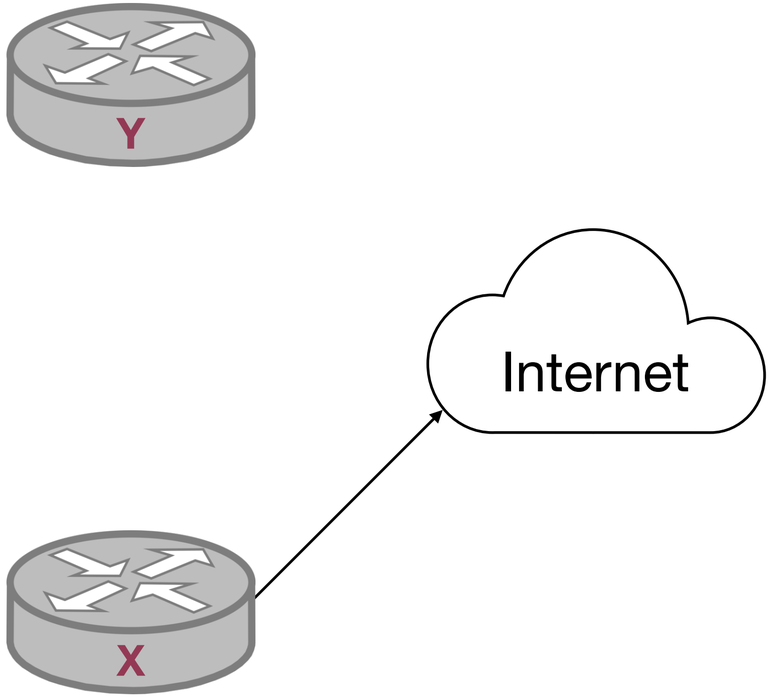


Are they all hijacks?

Question 1:
Why RPKI-Invalid Happens?

Known Misconfigurations



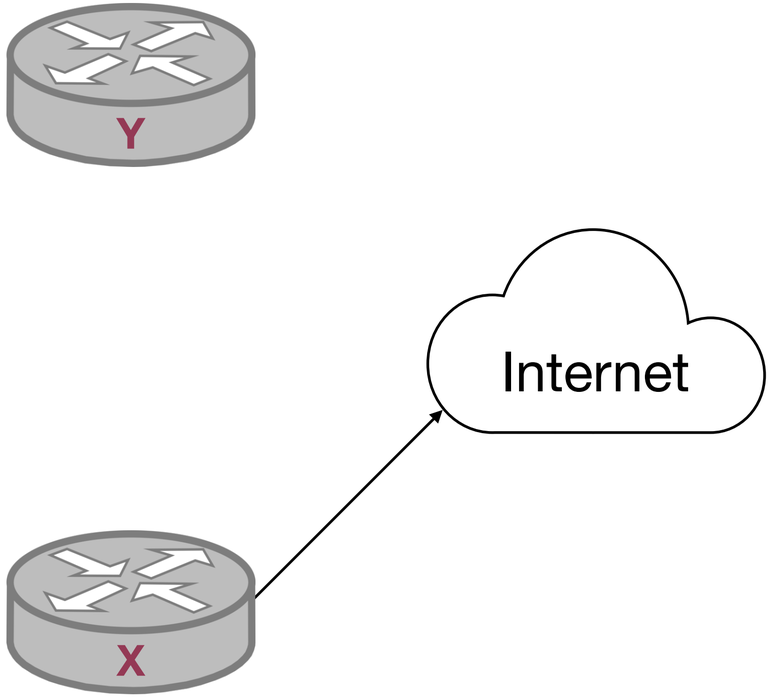
BGP Announcement:

129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

Known Misconfigurations



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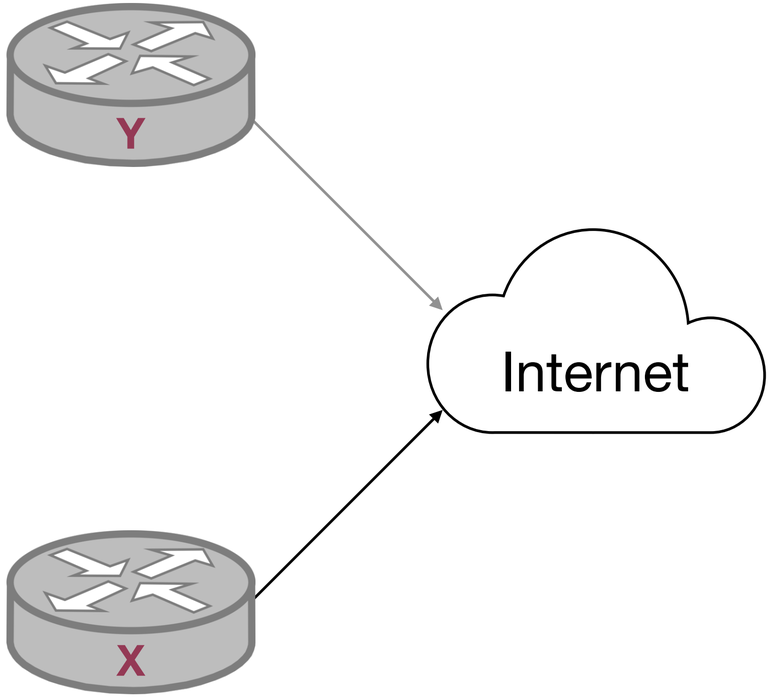

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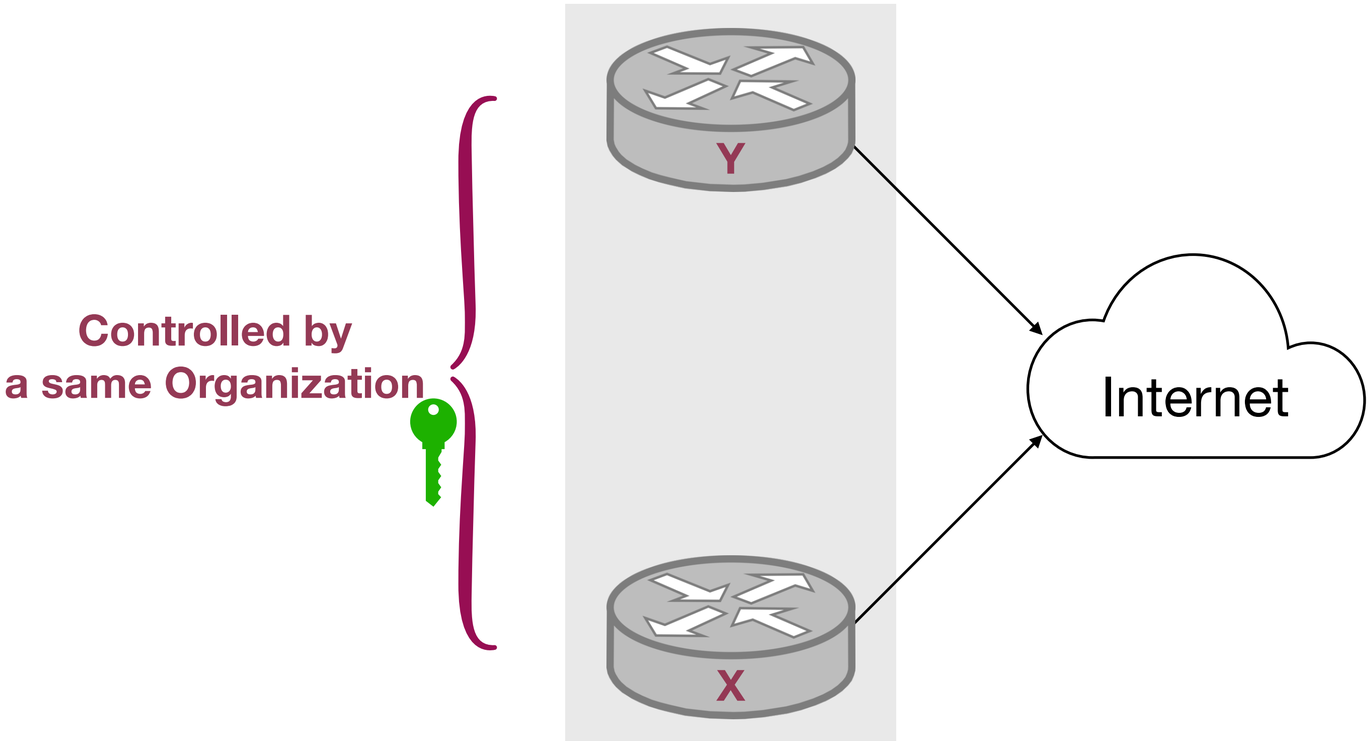
Known Misconfigurations in Same Org



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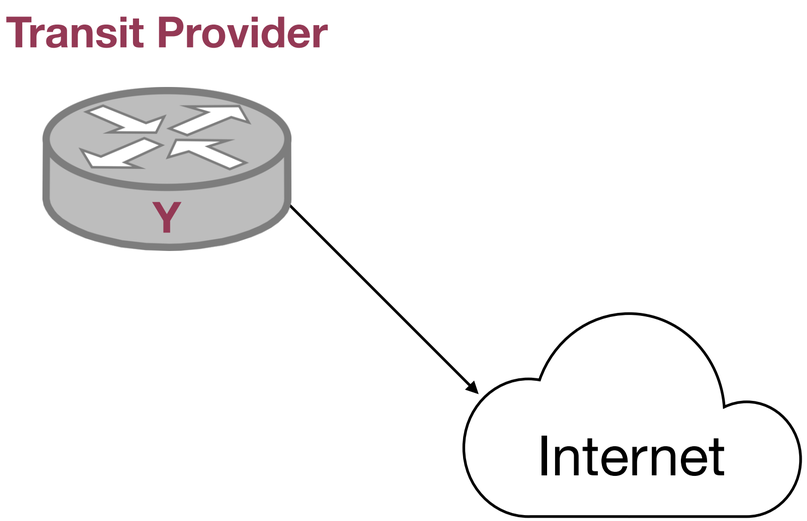
: Access to ROA

Known Misconfigurations in IP Transit



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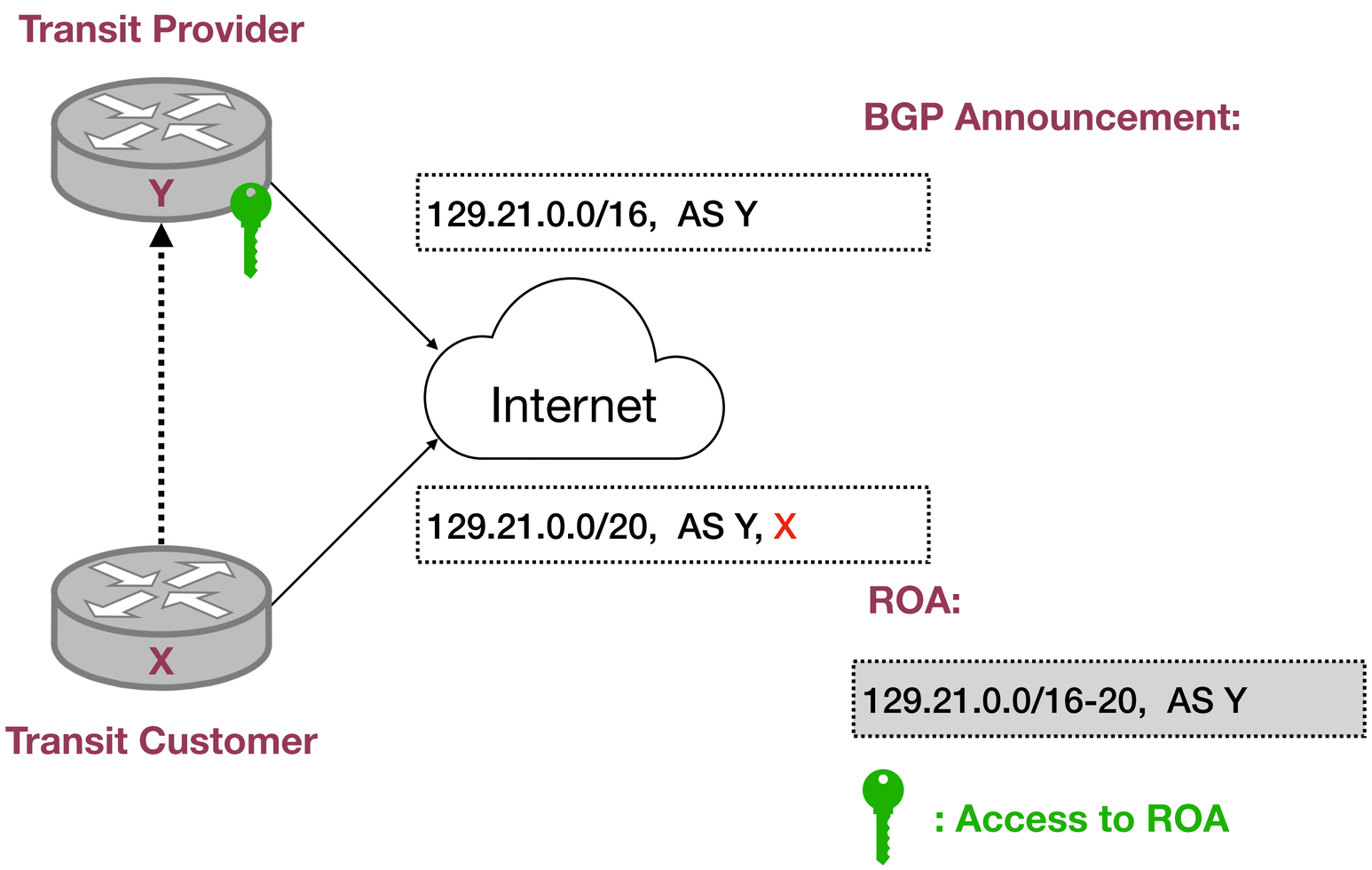

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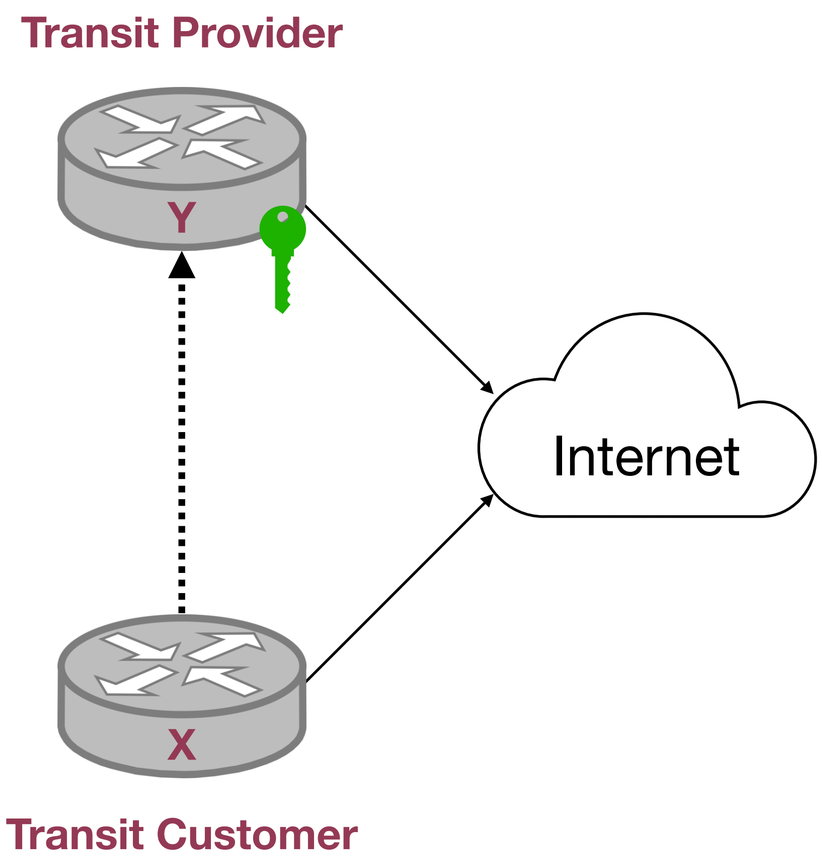

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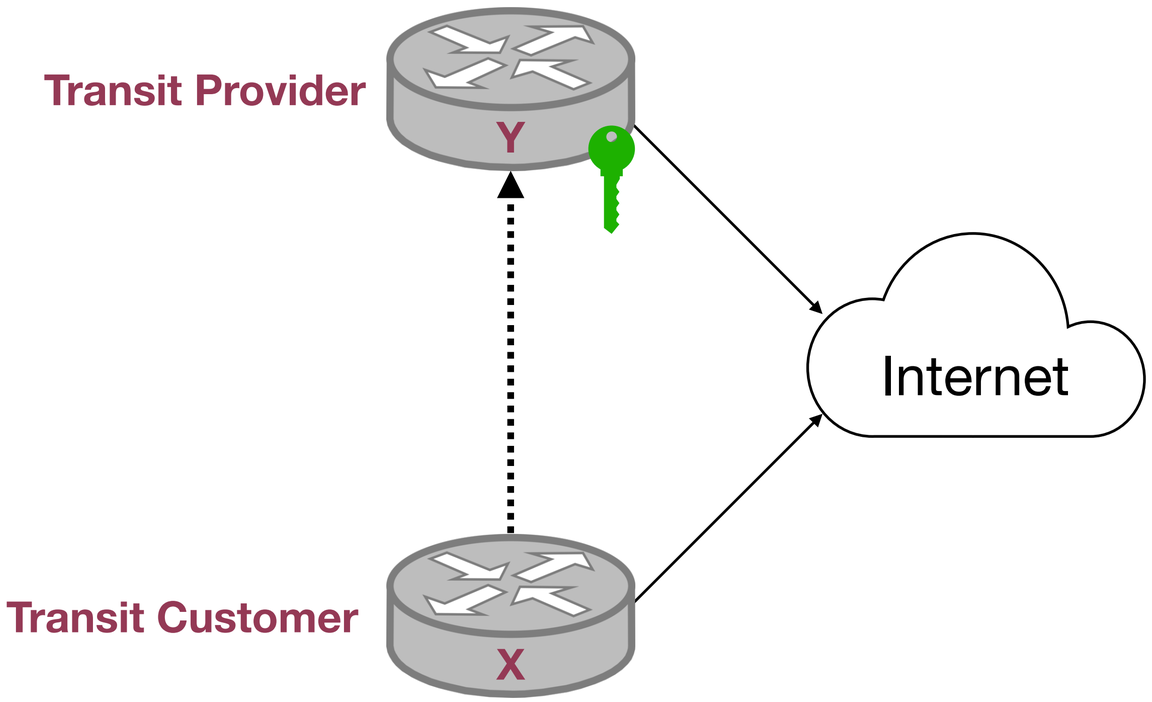

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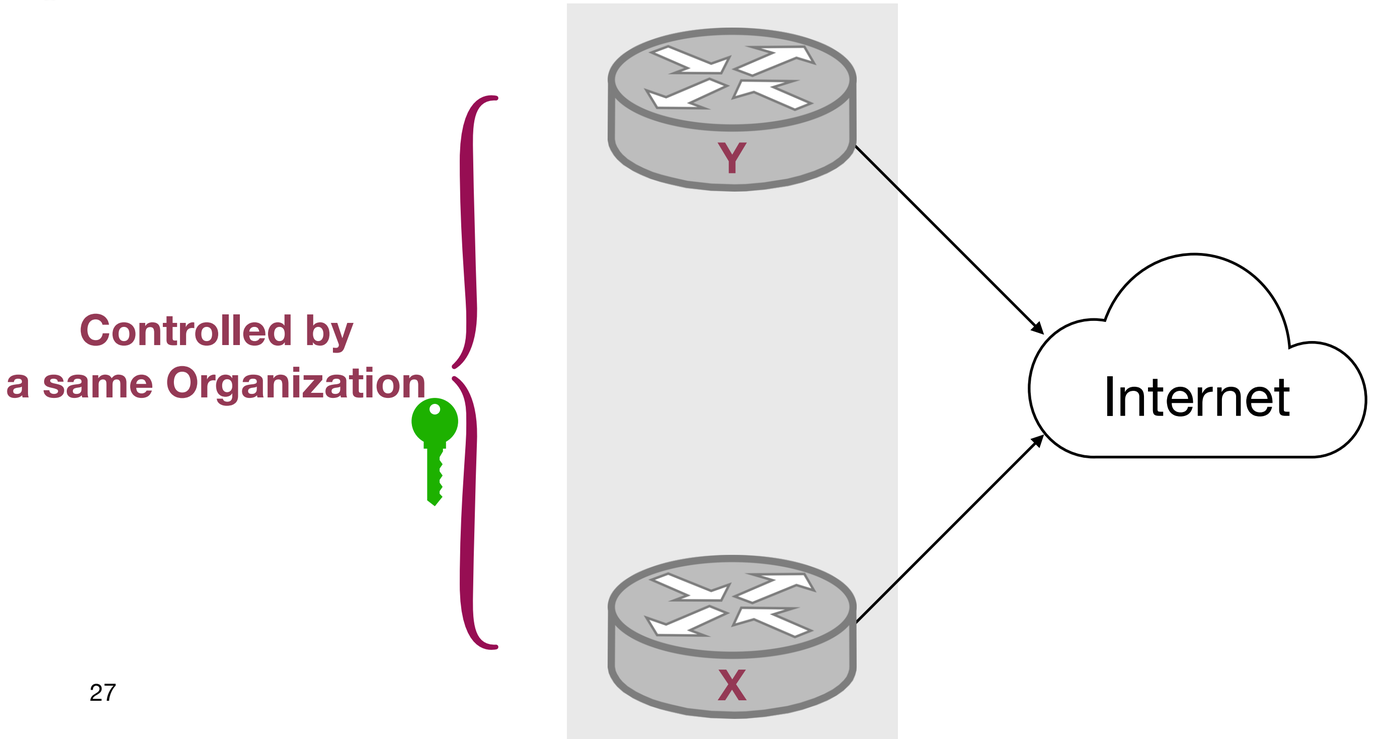

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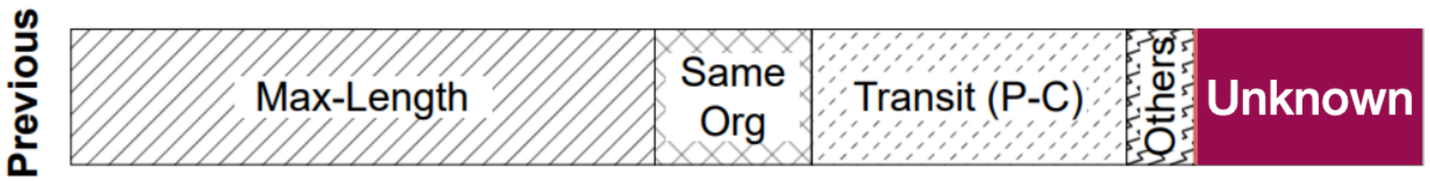

Are they all hijacks?

[1] IMC'19 RPKI is Coming of Age: A Longitudinal Study of RPKI Deployment and Invalid Route Origins

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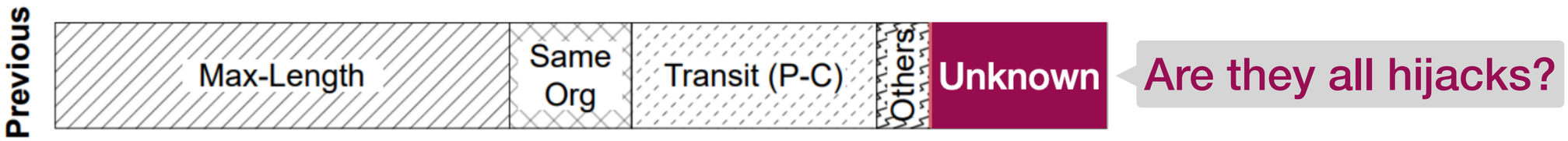

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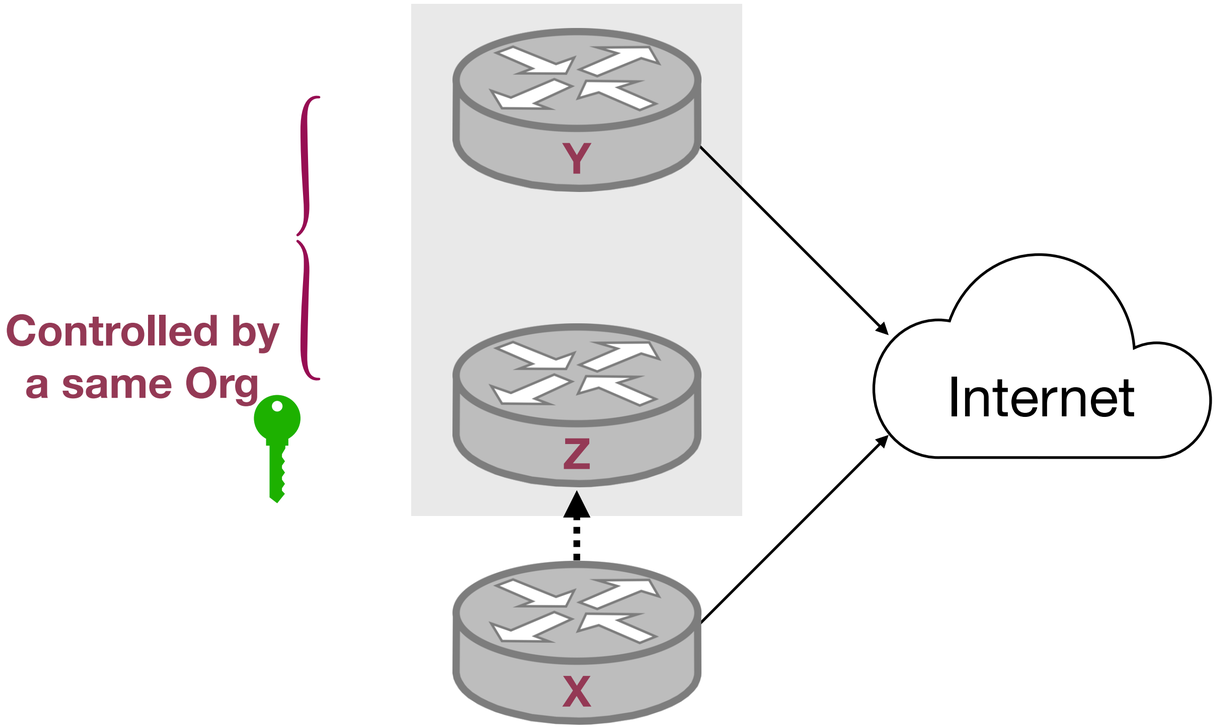
Hidden Relationships: Transit + Org



BGP Announcement:

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129.21.0.0/20, AS Z X

ROA:

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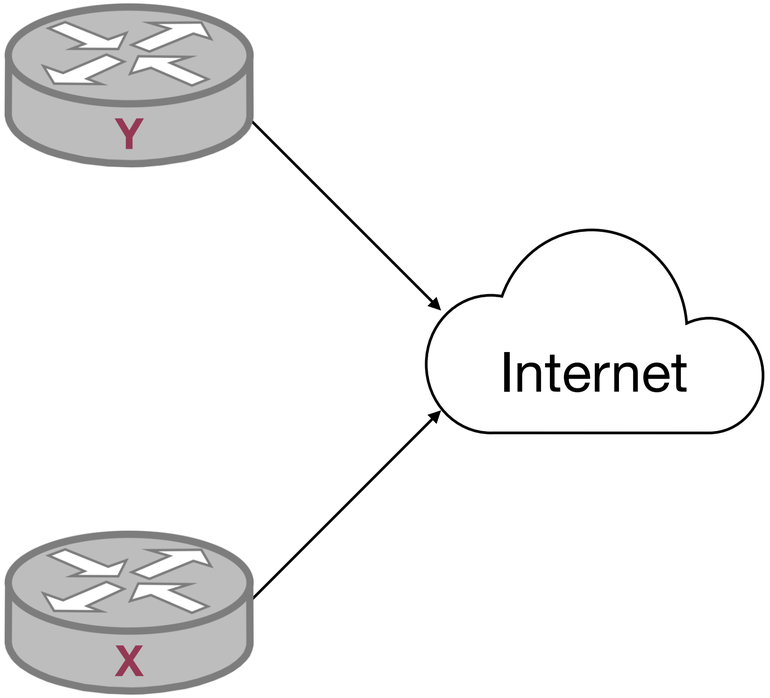
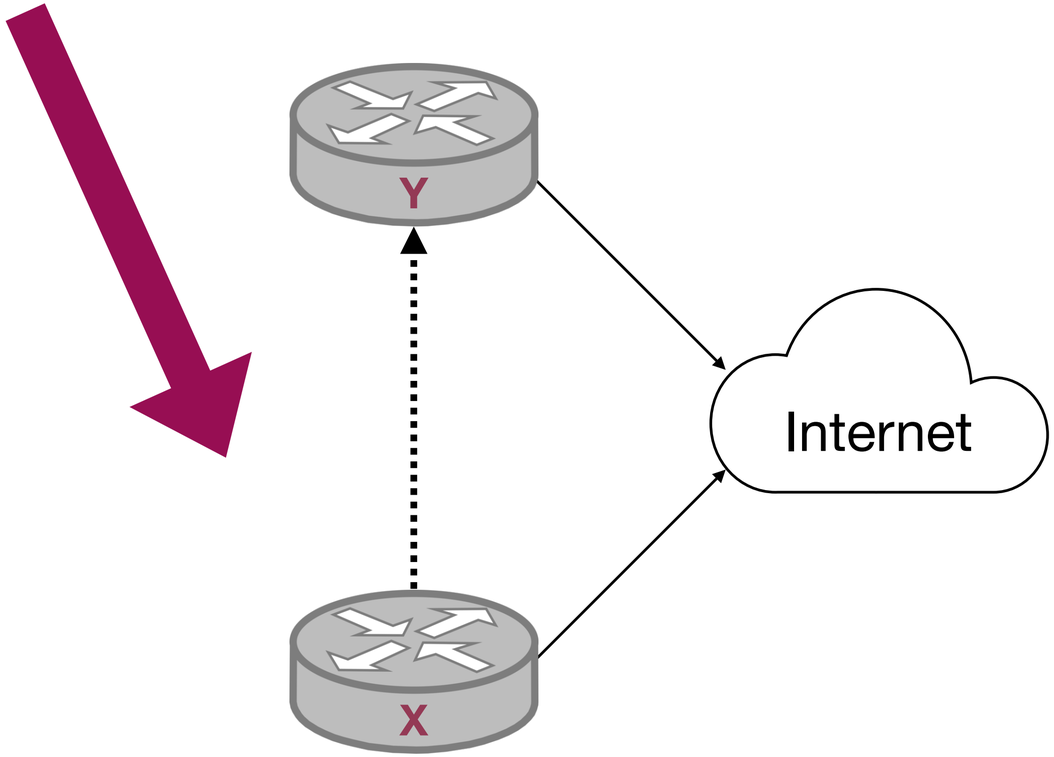
Hidden Relationships

- What if we can’t find any relationship between X and Y in BGP or AS2ORG mapping?



Hidden Relationships: Transit + Tunnel

Transit services may not show up in the BGP path!



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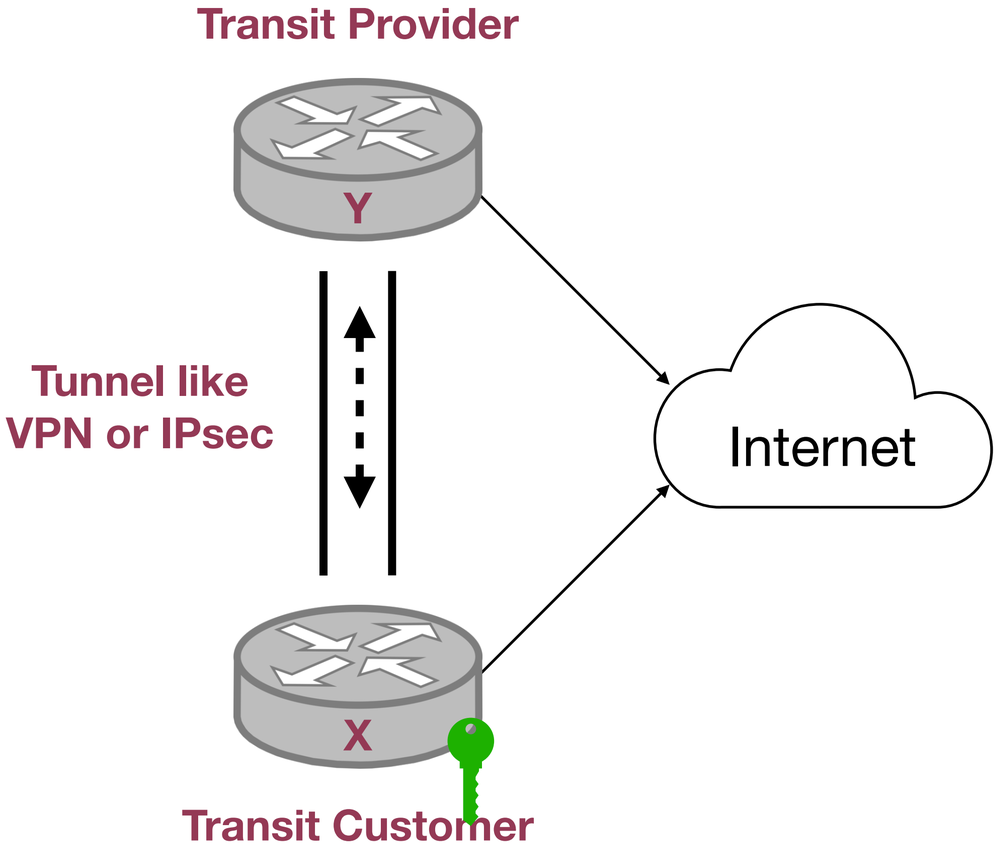

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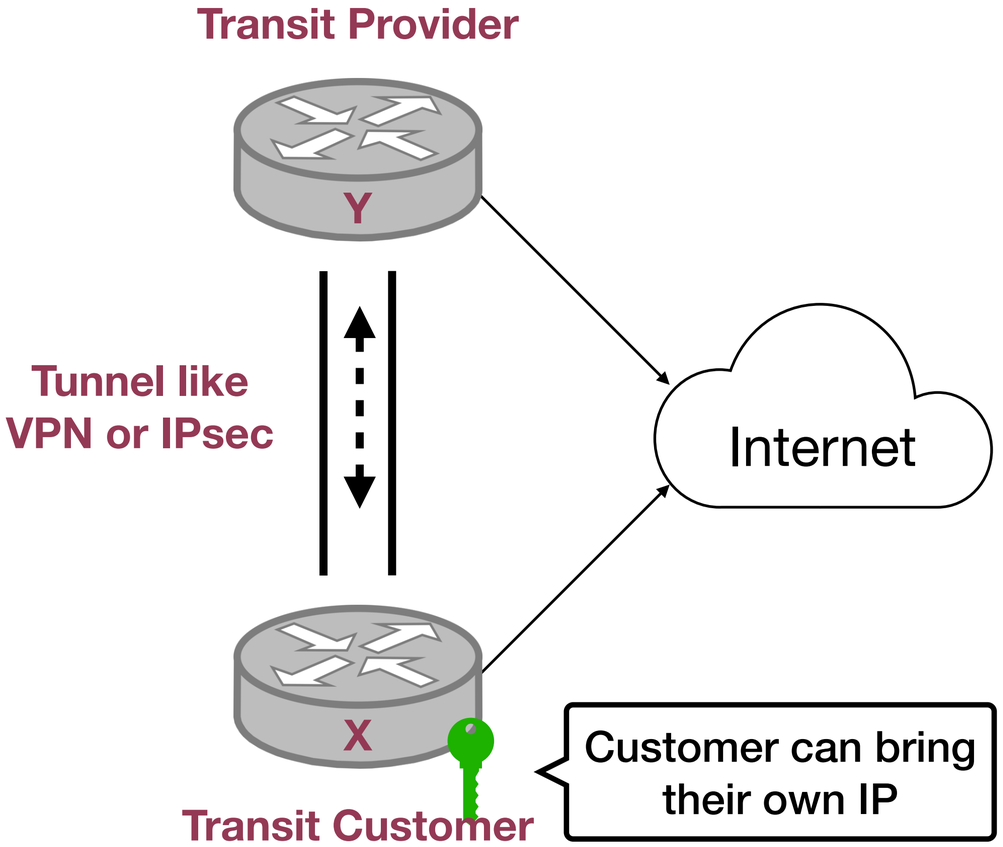

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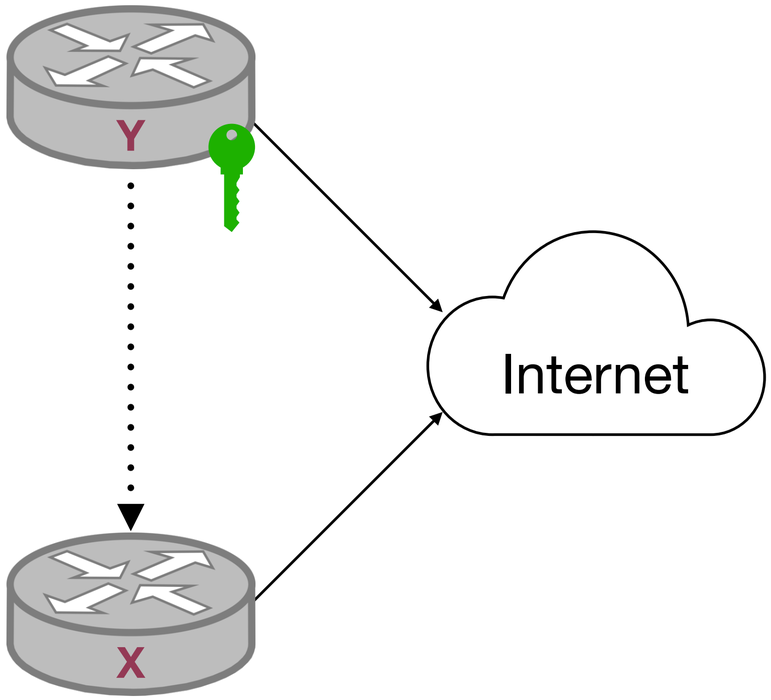
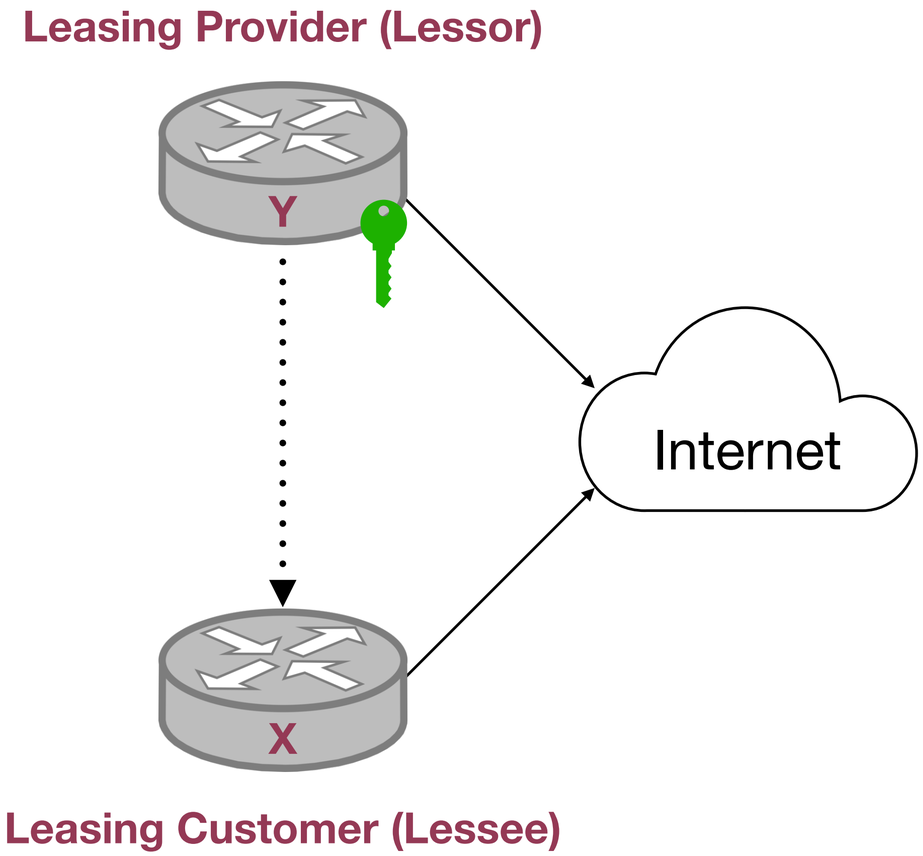
Hidden Relationships

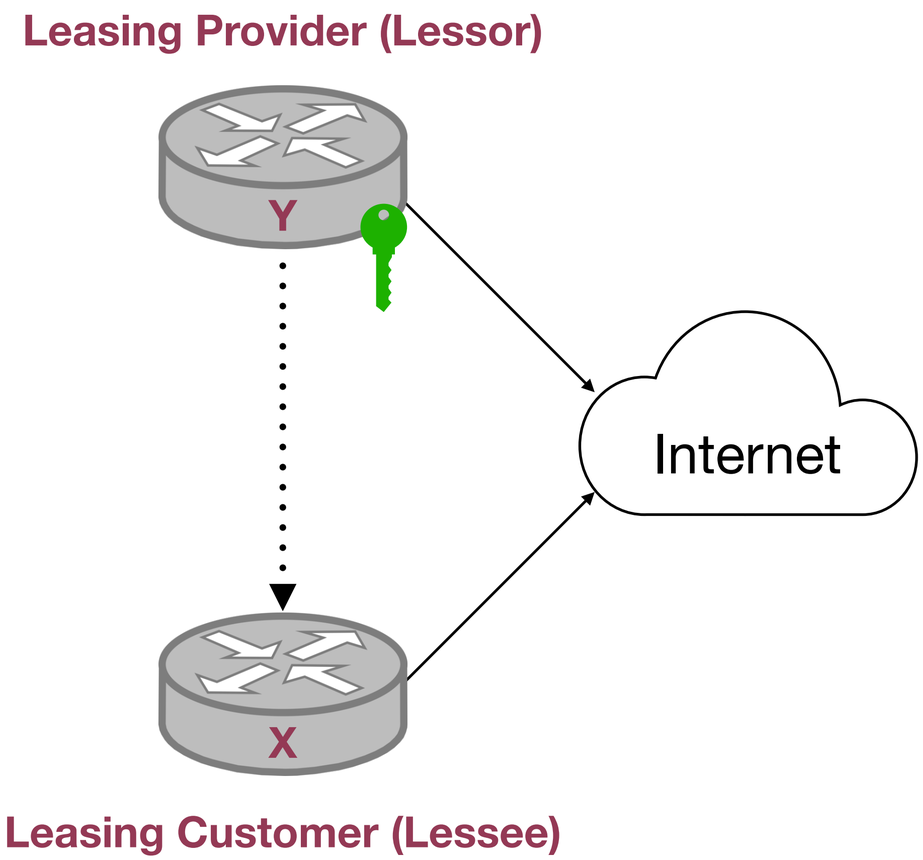
- Transit service is not the only reason!



Hidden Relationships: Leasing



Hidden Relationships: Leasing (direct)



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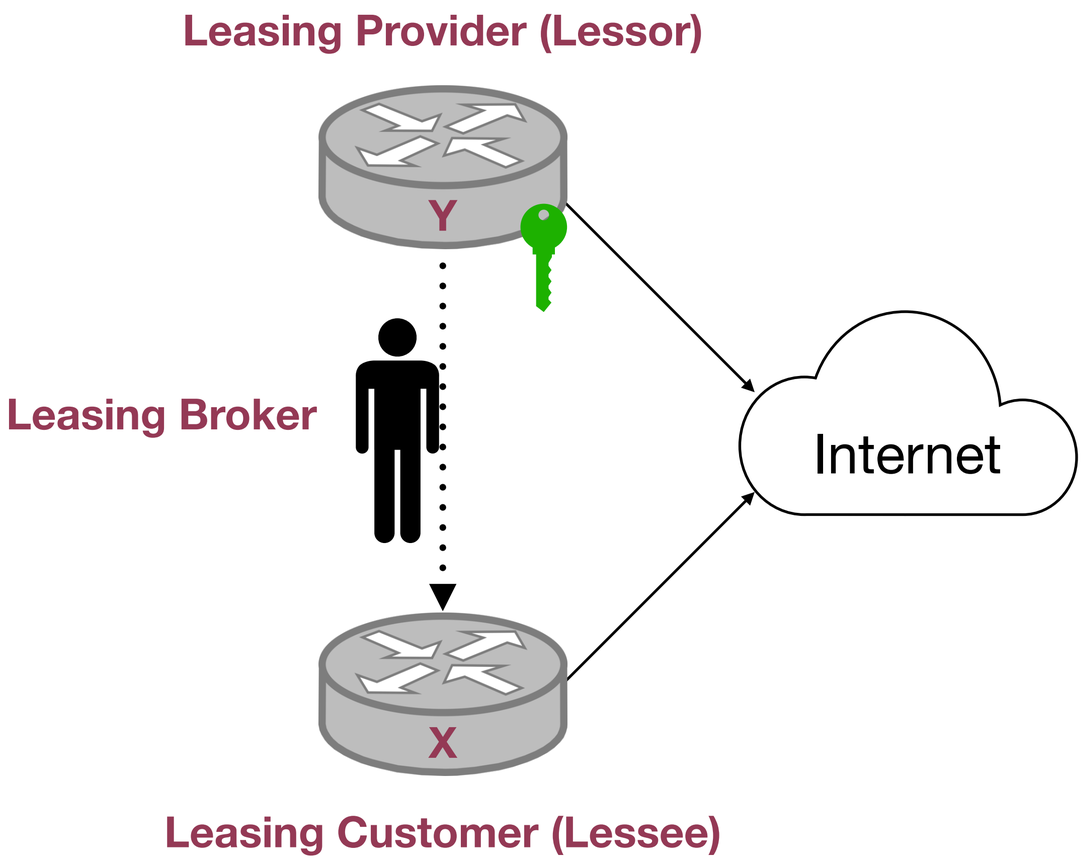
Hidden Relationships: Leasing (w/ Broker)



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Classification of RPKI-Invalid

Easy ones:

1. **Transit**: checking BGP path from BGP datasets from RouteViews and RIS
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4. **Broker leasing**: Broker usually will leave registration information in WHOIS

```
Customer:       IPXO LLC (C[illegible]061992)
RegDate:        2025-01-07
Updated:        [illegible]
Ref:            https://rdap.arin.net/registry/ip/67.43.36.0

CustName:       IPXO LLC
Address:        3132 State Street
City:           Dallas
StateProv:      TX
PostalCode:     75204-3500
Country:        US
RegDate:        2025-01-07
Updated:        2025-01-07
```

Classification for Hidden Relationships

Transit with tunneling and direct leasing are hard to identify:

- No evidence in BGP since the BGP origin will be the provider
- No evidence in WHOIS since the WHOIS record will remain lessor

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Thus we need to guess, but how?

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- Instead of trying to find individual prefix involved (which is hard to identify), we try to find the ASes who provide such hidden services (direct leasing and hidden transit).

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- Our hypothesis are:

  1. These service providers usually will have many customers

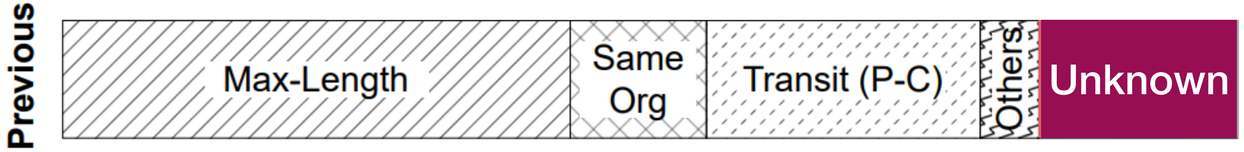
  2. Mistakes in ROA shouldn’t be a common situation

Classification for Hidden Relationships

- We collect all RPKI Valid prefixes from RouteViews and RIS BGP tables in the past 18 months.
- Identifying transit providers and lessors ASes and counting the involved prefix per AS
- Testing our methods with ground-truth leasing prefixes and BGP transit providers.

Classification Results

- We attribute 96.9% of the RPKI-Invalid prefixes to these misconfigurations



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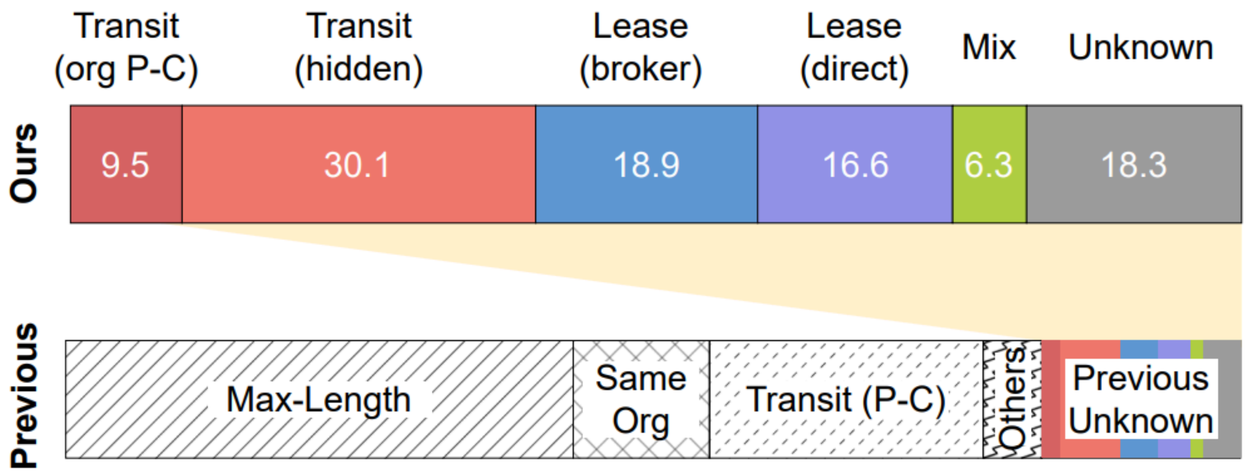
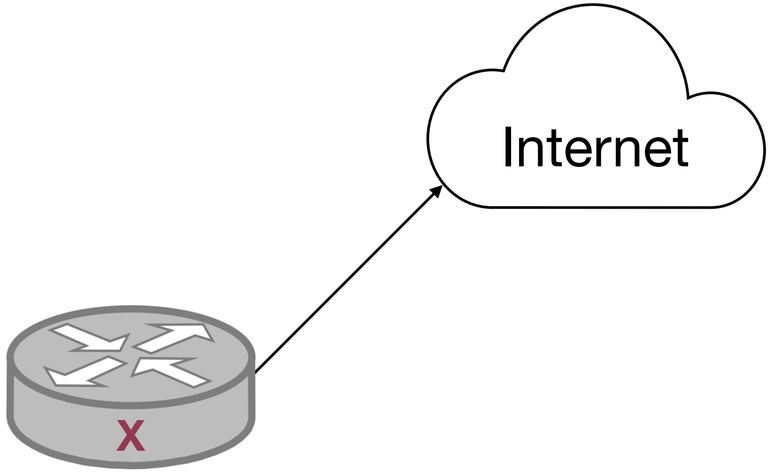


Question 2:
What’s the impact of RPKI-Invalid?

While people keep making mistakes in RPKI, it’s fine if there’s no impact on daily operation

But we find it’s not.

Impact of RPKI-Invalid: Disconnection



BGP Announcement:

129.21.0.0/20, AS X

ROA:

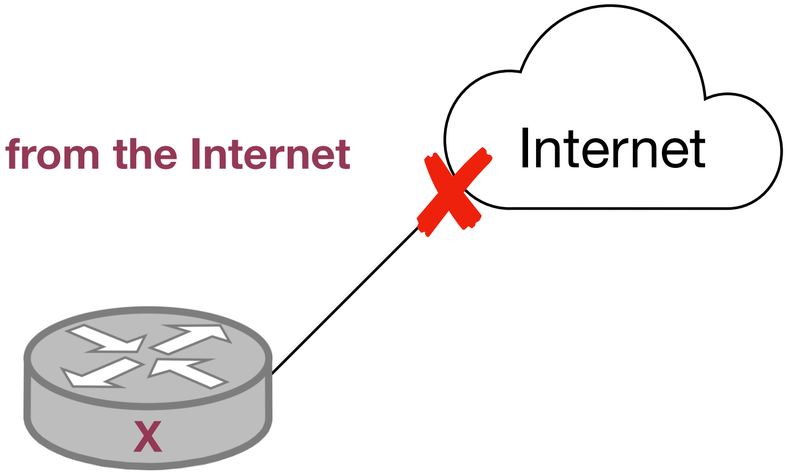
129.21.0.0/16-20, AS Y

Impact of RPKI-Invalid: Disconnection

BGP Announcement:

129.21.0.0/20, AS X

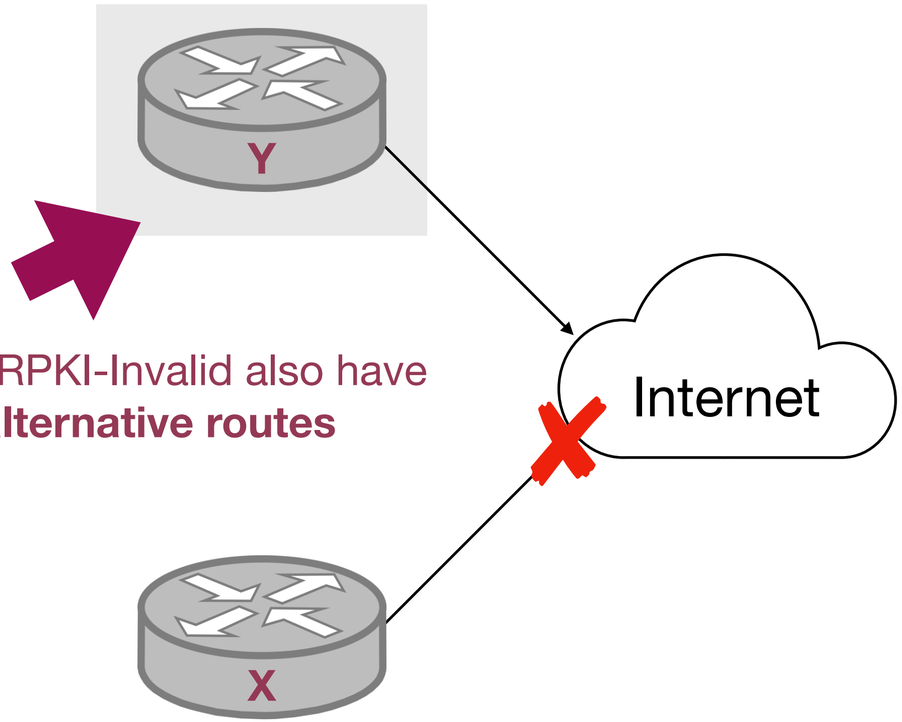
Disconnected from the Internet



ROA:

129.21.0.0/16-20, AS Y

Impact of RPKI-Invalid: Disconnection



But, more than 95% RPKI-Invalid also have valid/unknown **alternative routes**

BGP Announcement:

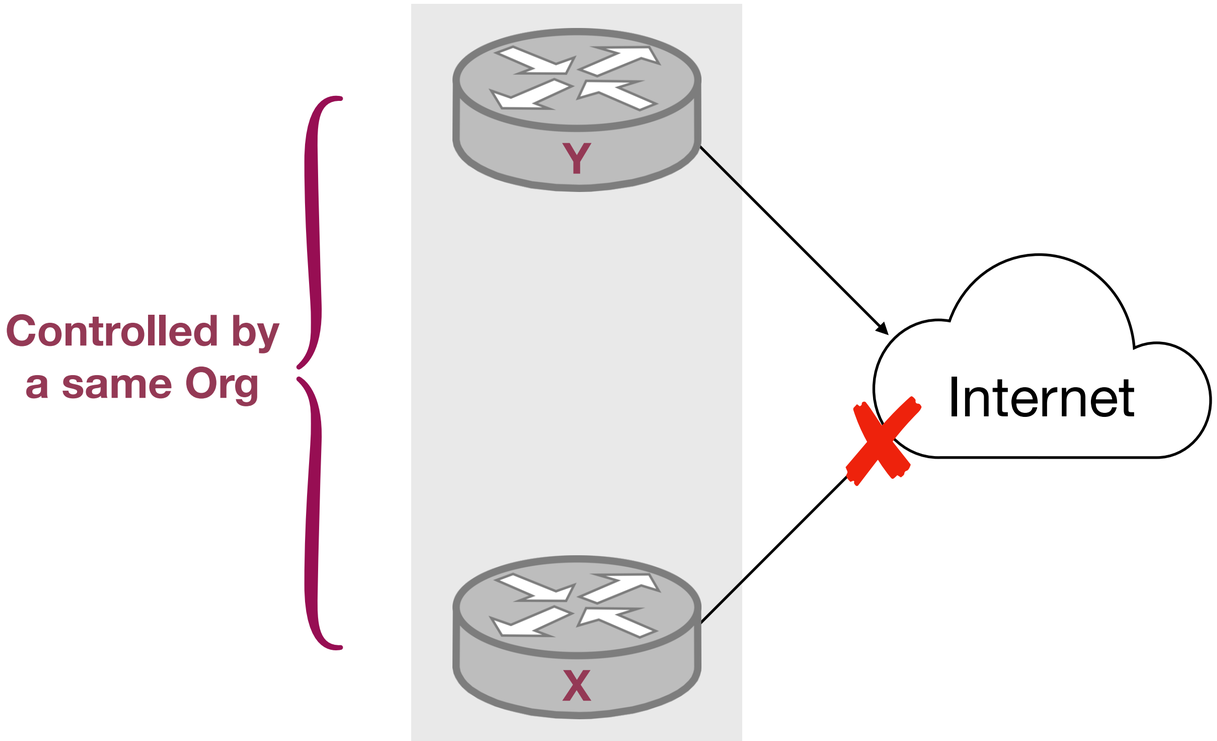
129.21.0.0/16, AS Y

129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

Impact of RPKI-Invalid: Disconnection



BGP Announcement:

129.21.0.0/16, AS Y

129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

Impact of RPKI-Invalid: Disconnection

- We run active measurements, using RIPE Atlas probes to ping&traceroute all invalid prefixes.

Impact of RPKI-Invalid: Disconnection

- We run active measurements, using RIPE Atlas probes to ping&traceroute all invalid prefixes.
- We found only 3.1% of RPKI-Invalid will result to disconnection.

	Disconnection (%)		
	0	0-25	25-100
Total	96.9	2.0	1.1
Max Length	97.1	1.9	1.0
Same ORG	96.7	2.3	1.0
P-C Transit	98.3	1.5	0.2
Org Level Transit	98.0	1.5	0.5
Hidden Transit	97.6	1.8	0.6
Direct Leasing	95.5	3.1	1.3
Broker Leasing	94.3	4.3	2.0
Leasing + Transit	95.2	3.7	1.8
Unknown	91.6	5.0	2.6

Impact of RPKI-Invalid

- However, having alternative path doesn’t means no impact at all

Impact of RPKI-Invalid

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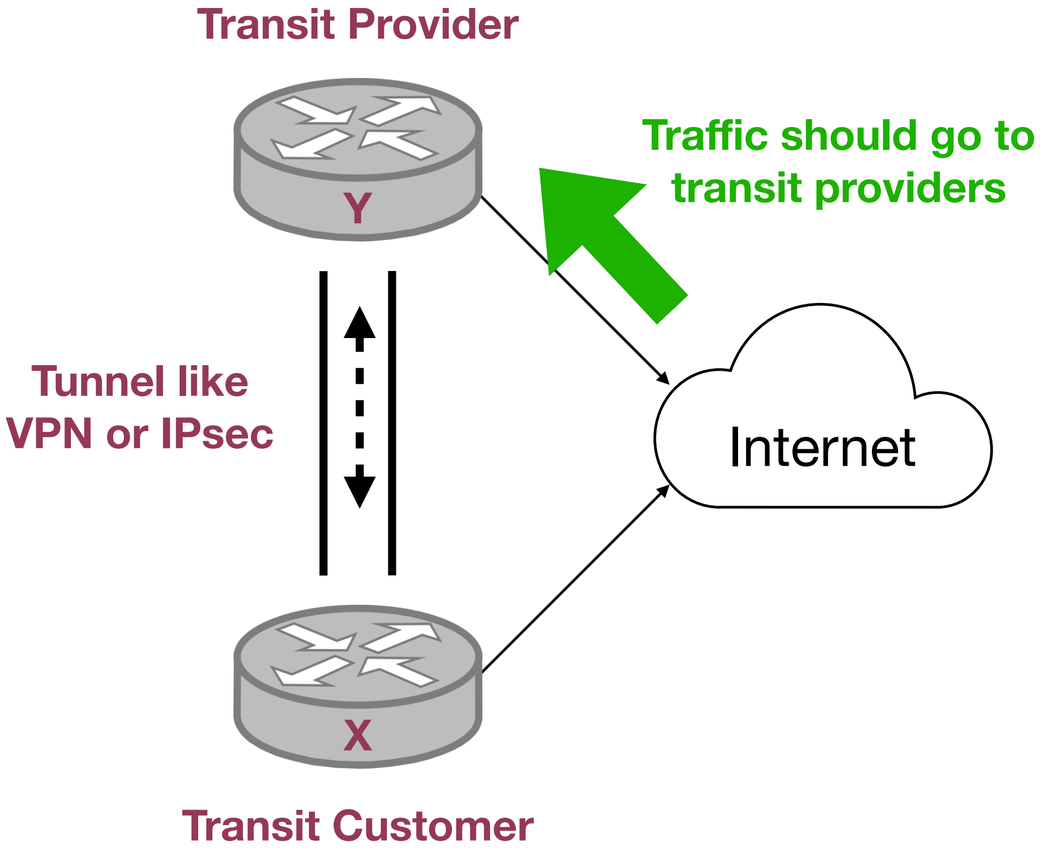

BGP Announcement:

129.21.0.0/16, AS X

129.21.0.0/20, AS Y

ROA:

129.21.0.0/16-20, AS X

Impact of RPKI-Invalid

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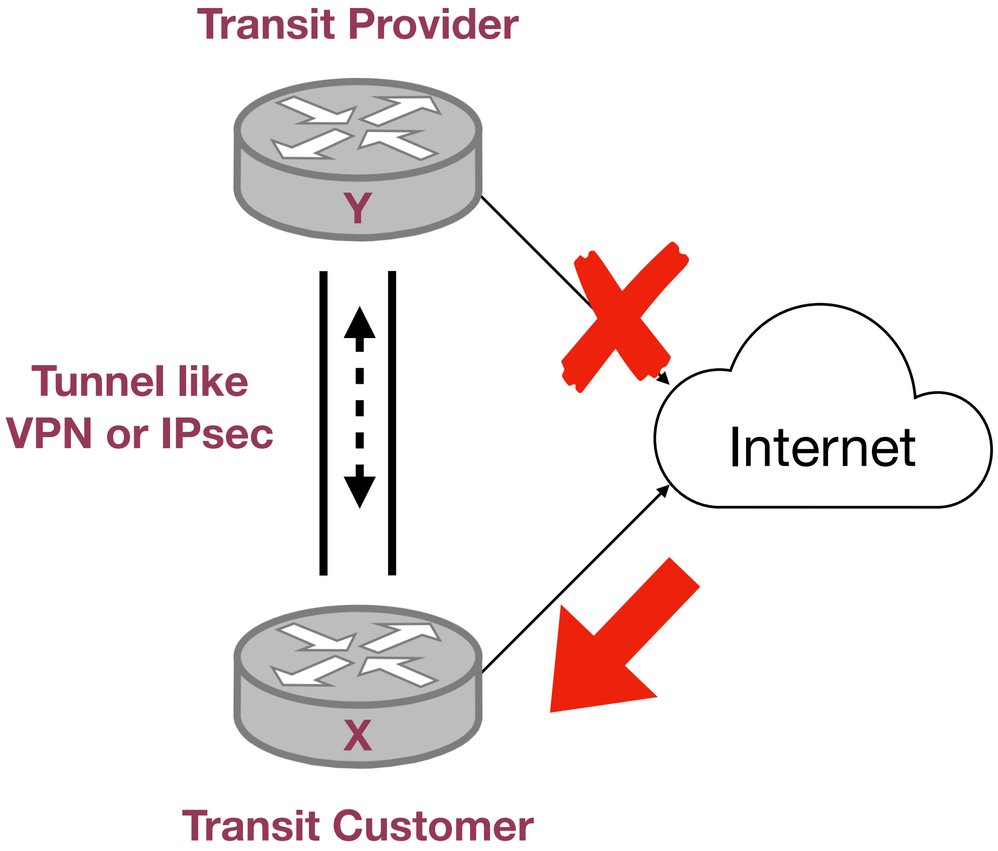


BGP Announcement:

129.21.0.0/16, AS X

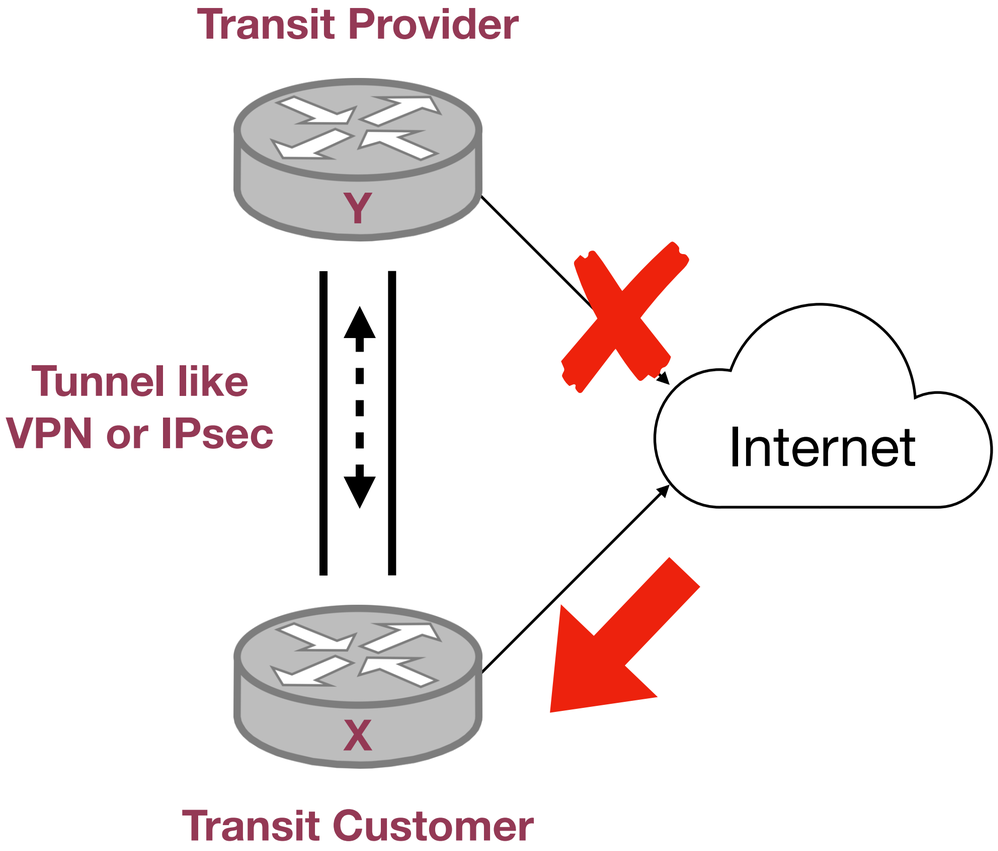
129.21.0.0/20, AS Y

ROA:

129.21.0.0/16-20, AS X

Impact of RPKI-Invalid: Path Divergence

- However, having alternative path doesn't means no impact at all.



BGP Announcement:

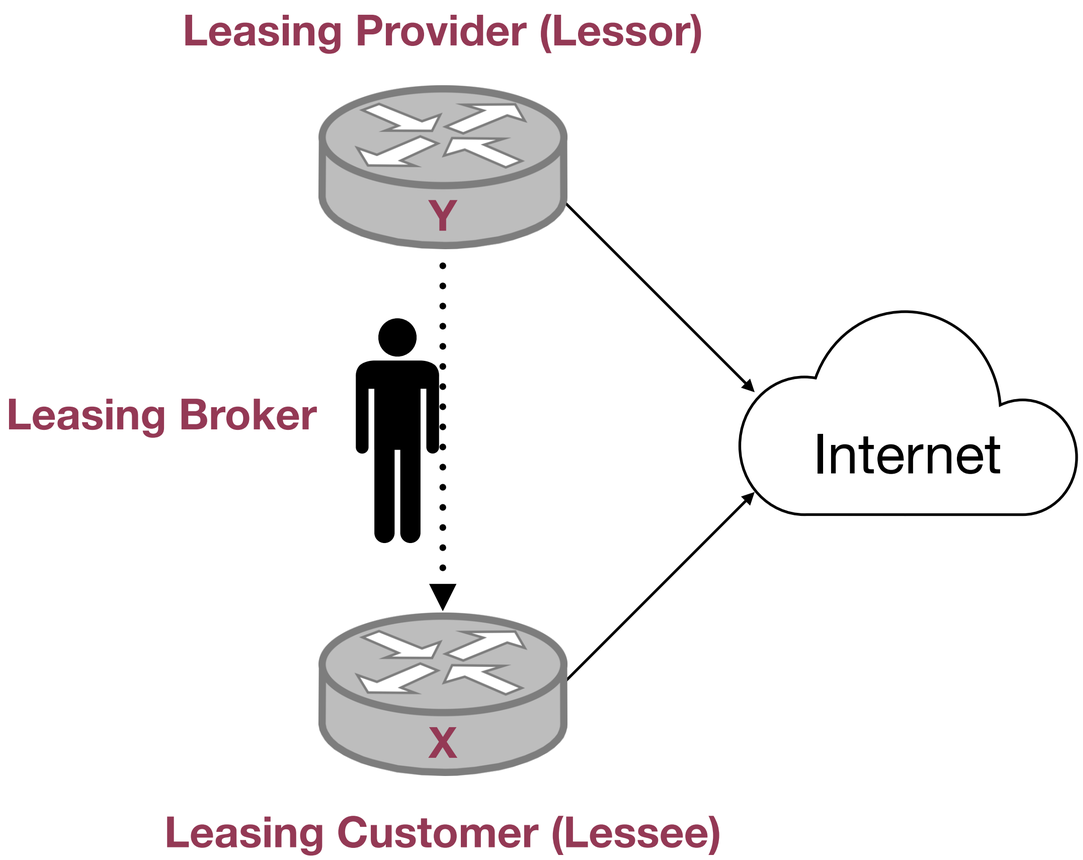
129.21.0.0/16, AS X

129.21.0.0/20, AS Y

ROA:

129.21.0.0/16-20, AS X

Impact of RPKI-Invalid: Path Divergence



BGP Announcement:

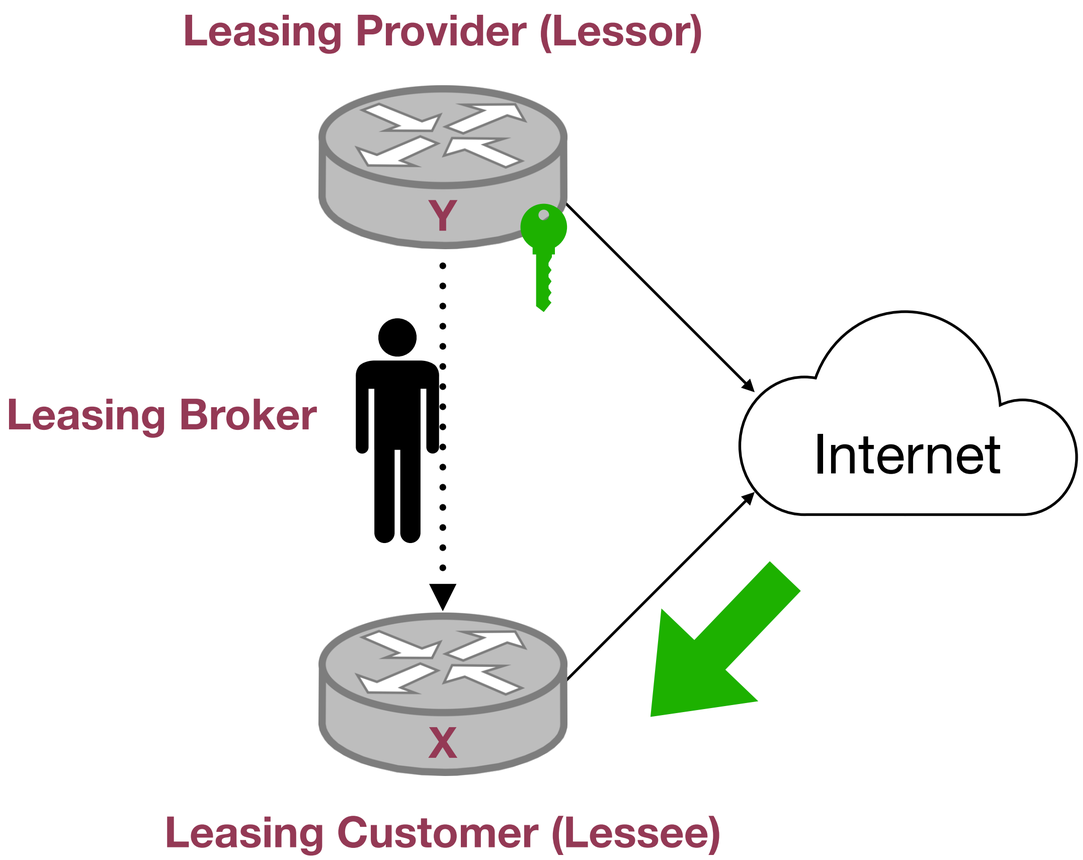
129.21.0.0/16, AS Y

129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

Impact of RPKI-Invalid: Path Divergence



BGP Announcement:

129.21.0.0/16, AS Y

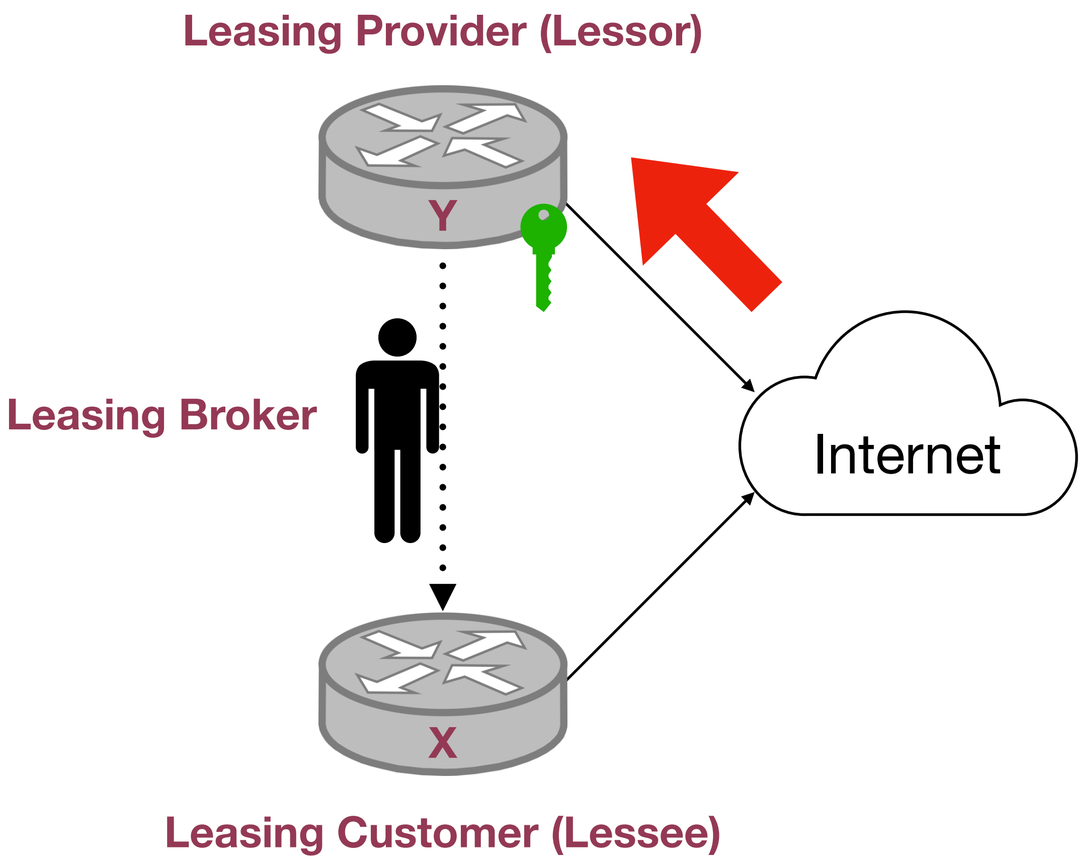
129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

: Access to ROA

Impact of RPKI-Invalid: Path Divergence



BGP Announcement:

129.21.0.0/16, AS Y

129.21.0.0/20, AS X

ROA:

129.21.0.0/16-20, AS Y

: Access to ROA

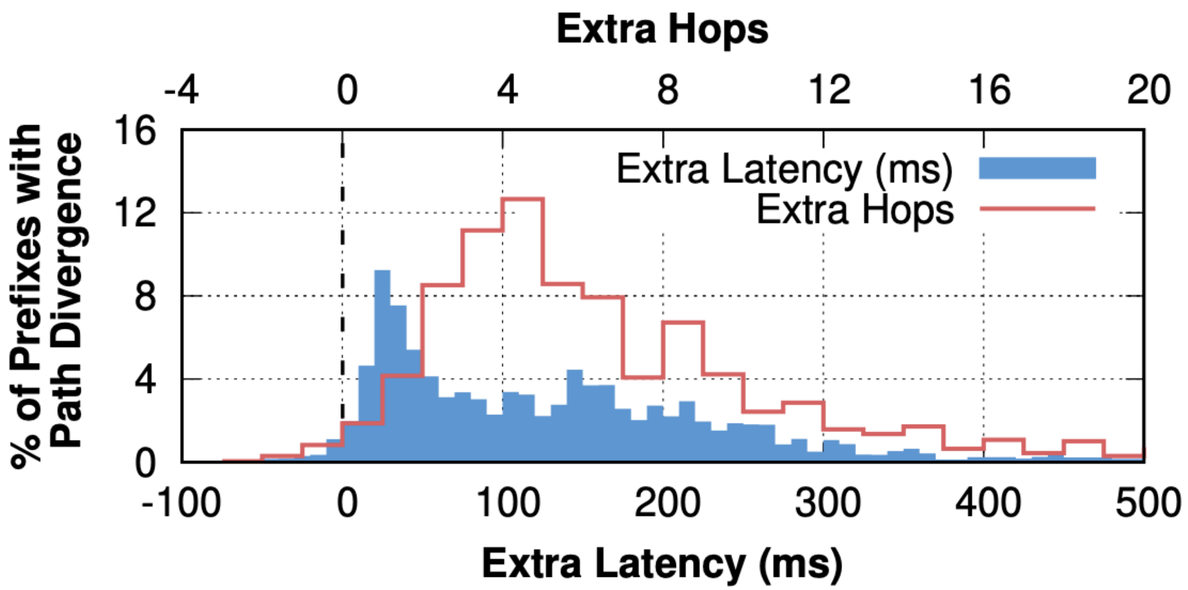
Impact of RPKI-Invalid: Path Divergence

- Path divergence is more common (18.5%) compared to disconnection

	Disconnection (%)			Path Divergence (%)		
	0	0-25	25-100	0	0-25	25-100
Total	96.9	2.0	1.1	81.5	12.1	6.4
Max Length	97.1	1.9	1.0	-	-	-
Same ORG	96.7	2.3	1.0	-	-	-
P-C Transit	98.3	1.5	0.2	89.6	7.6	2.8
Org Level Transit	98.0	1.5	0.5	74.2	15.1	10.7
Hidden Transit	97.6	1.8	0.6	70.0	19.4	10.6
Direct Leasing	95.5	3.1	1.3	68.5	21.5	10.0
Broker Leasing	94.3	4.3	2.0	65.1	23.3	11.6
Leasing + Transit	95.2	3.7	1.8	62.3	20.2	17.5
Unknown	91.6	5.0	2.6	63.9	22.8	13.3

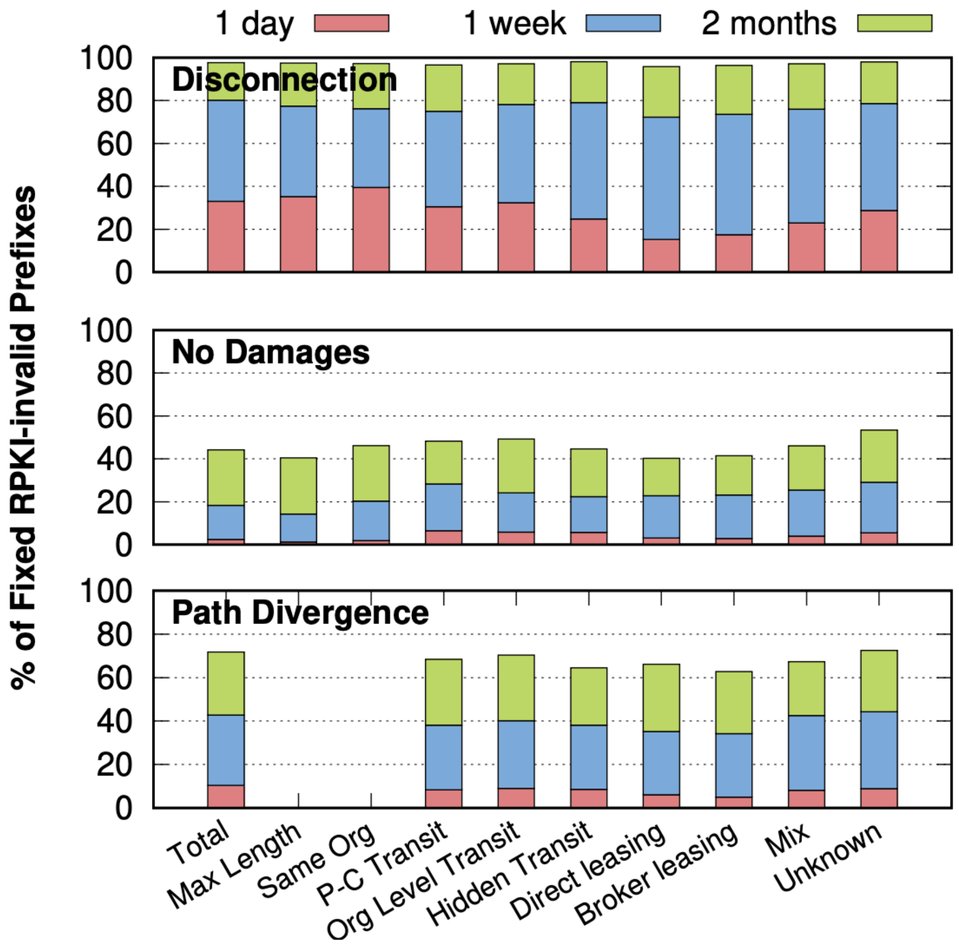
Impact of RPKI-Invalid: Path Divergence

- Path divergence is more common (18.5%) compared to disconnection
- Leading to additional hops, impact performance and protection, and increases the risk of MITM



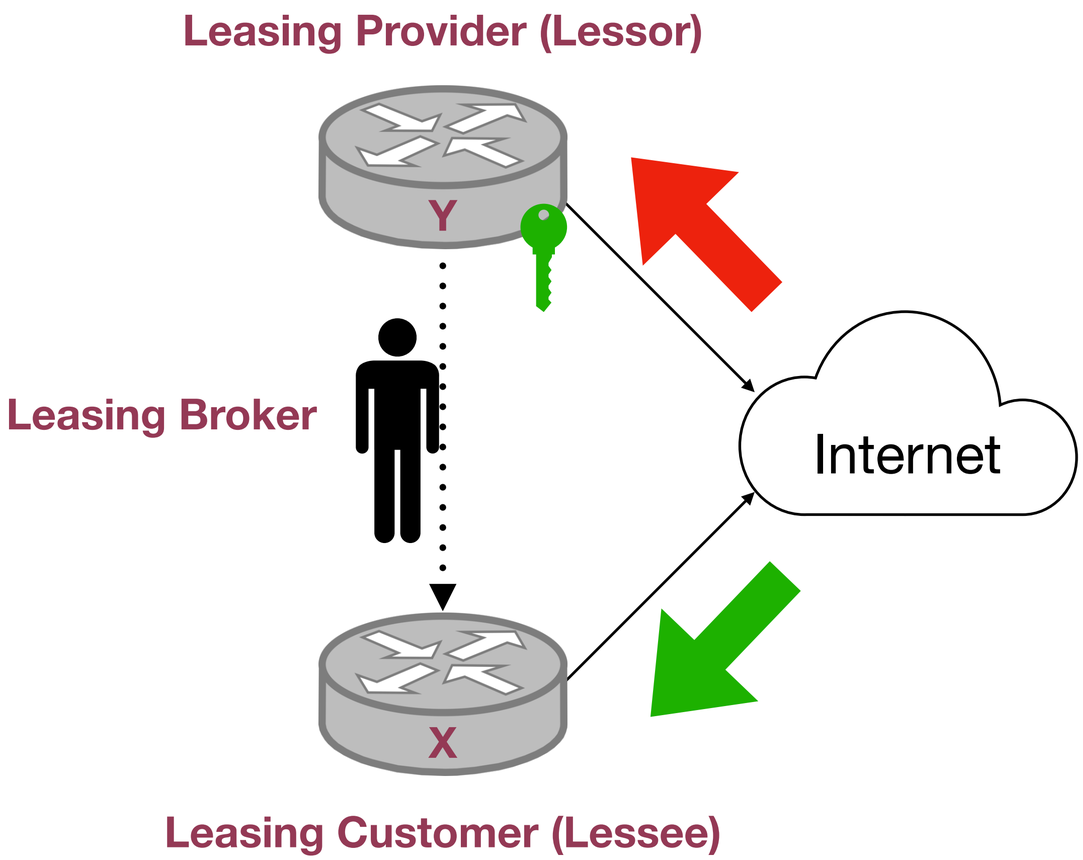
Fixing of RPKI-Invalid

- RPKI-Invalid prefixes are fixed quickly only if they result in disconnection



Question 3:
Can this be explored by an adversary?

Impact of RPKI-Invalid: Path Divergence



BGP Announcement:

129.21.0.0/16, AS Y

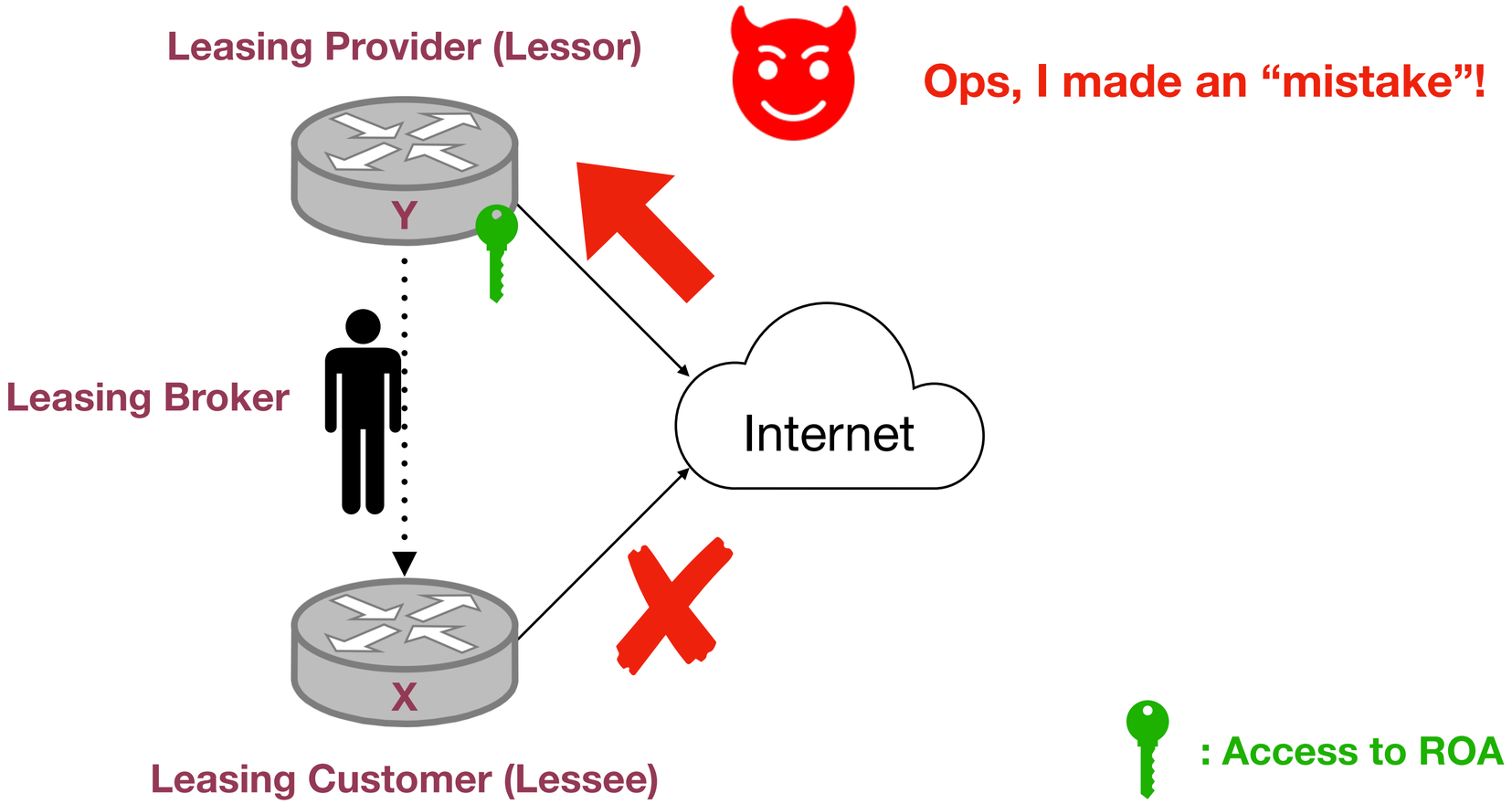
129.21.0.0/20, AS X

ROA:

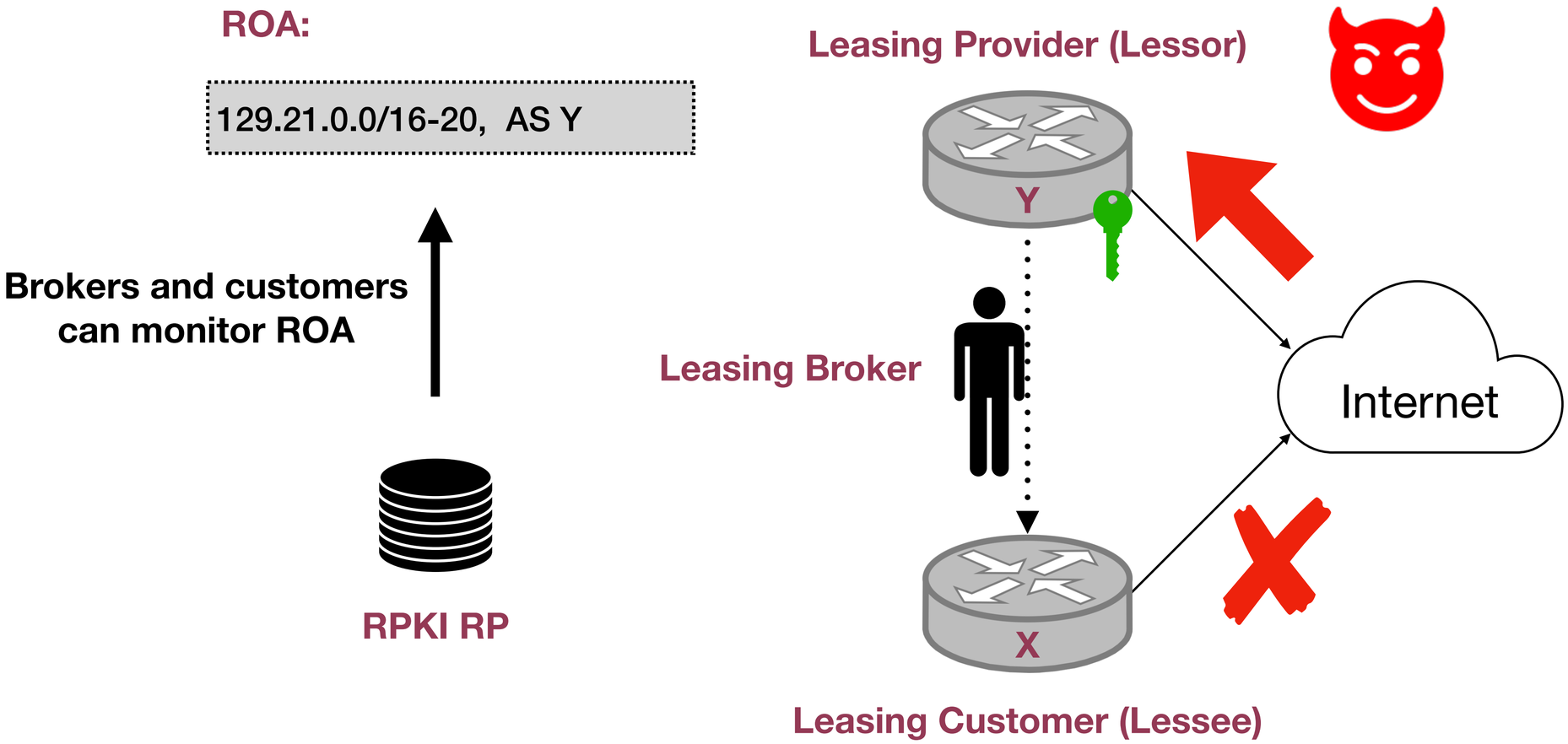
129.21.0.0/16-20, AS Y

: Access to ROA

A Rogue Lessor Can Explore RPKI



Brokers and Customers Can Monitor ROA



A Rogue Lessor Can Evade Monitoring

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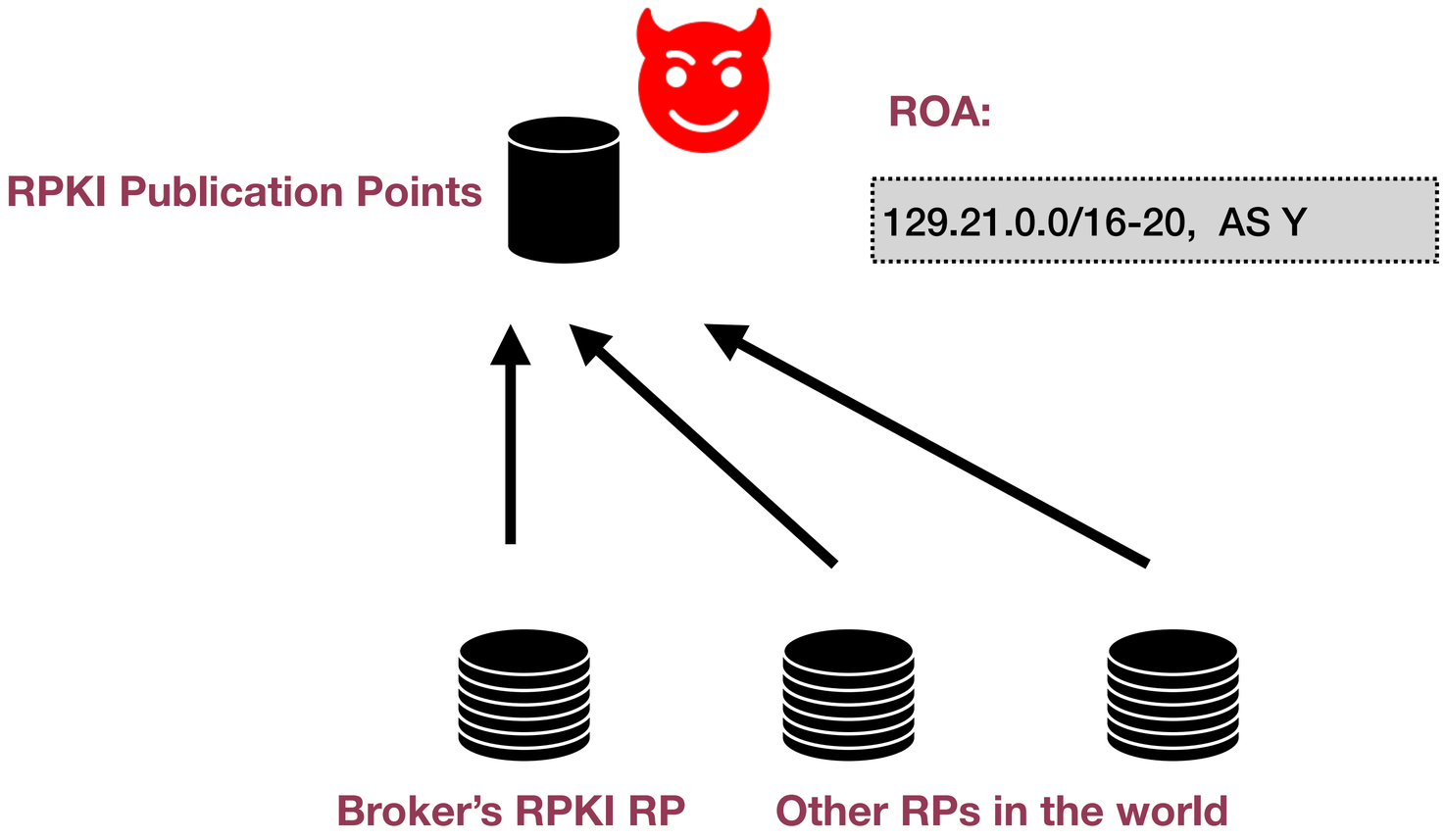

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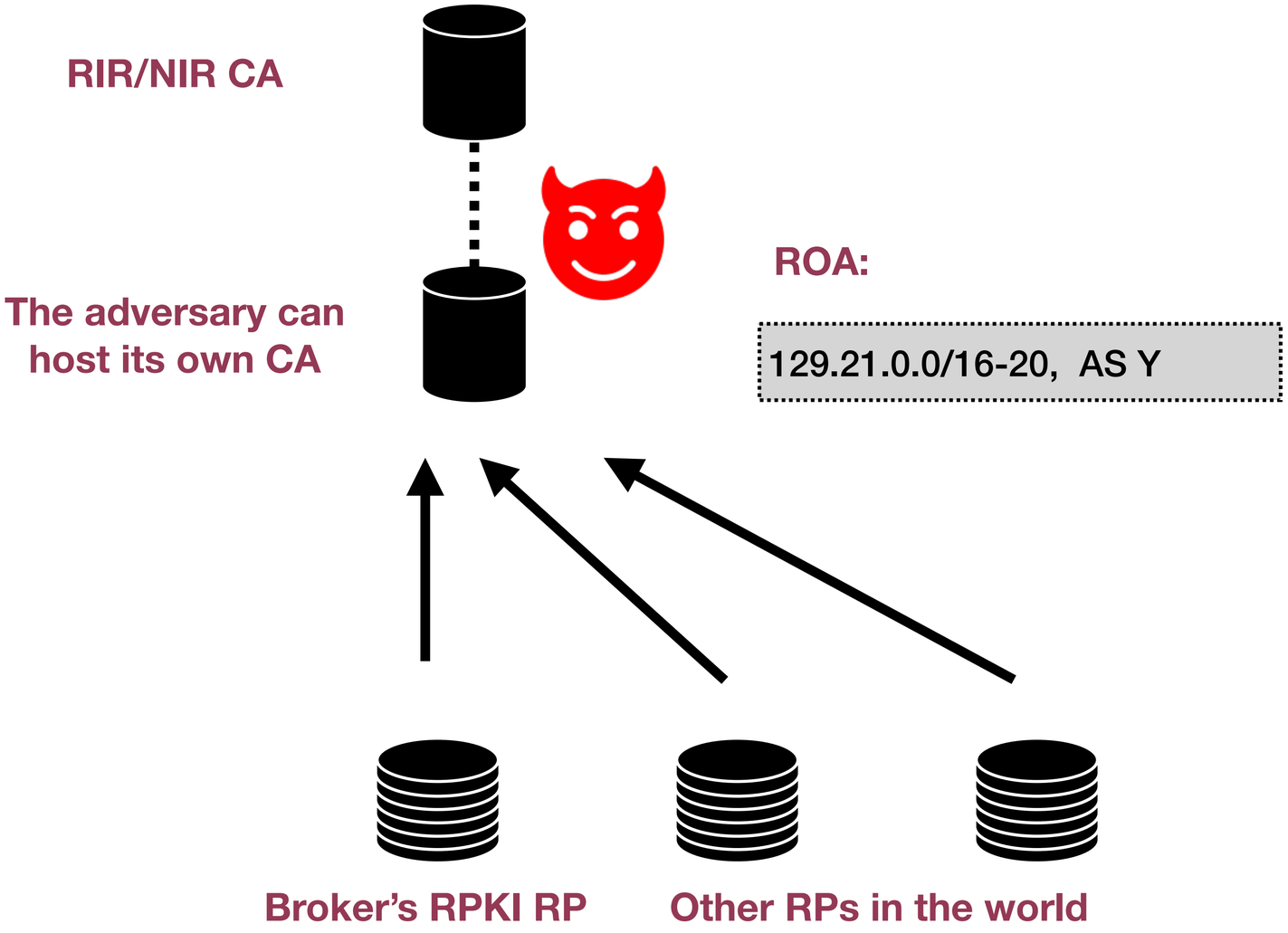

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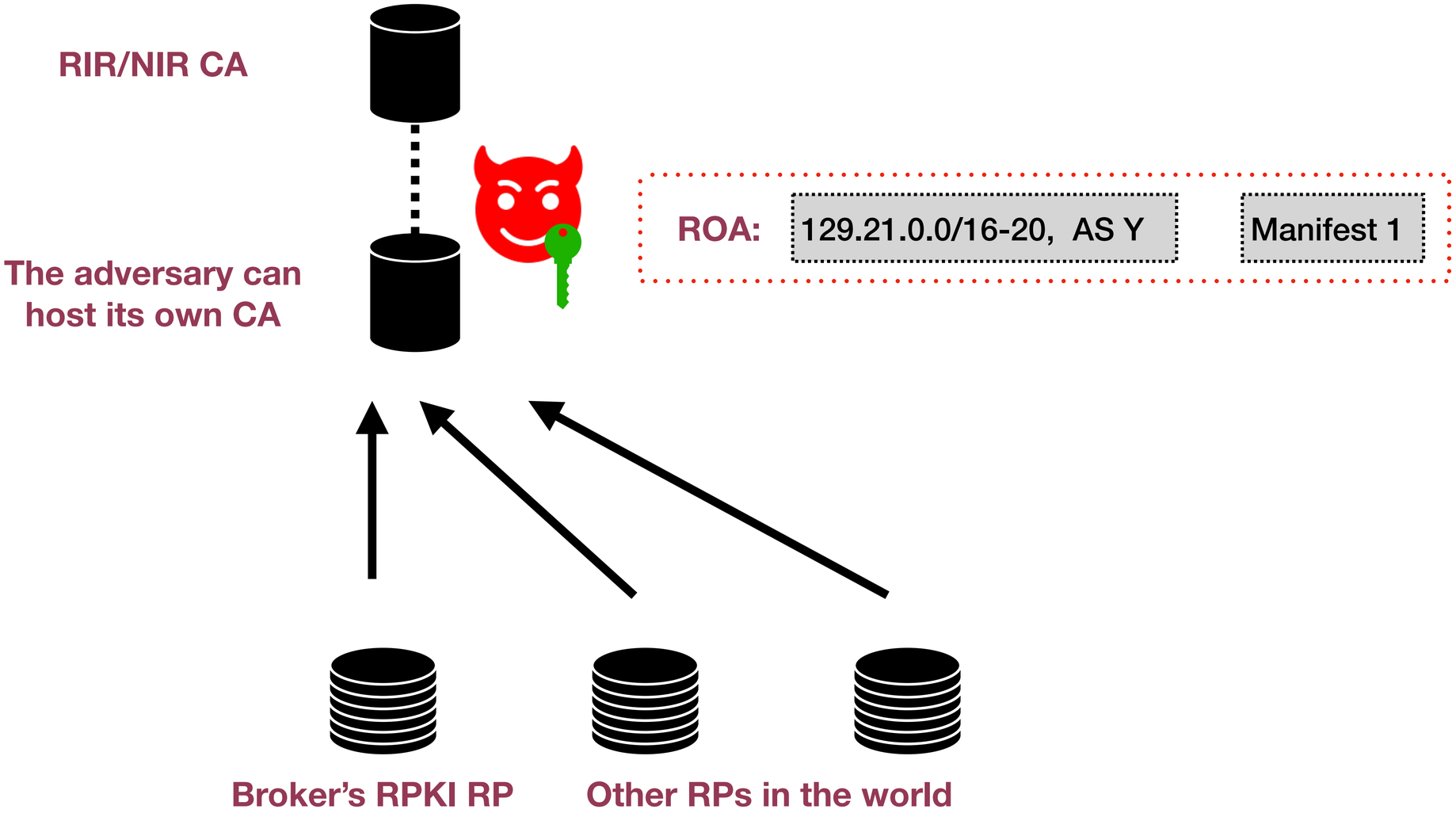

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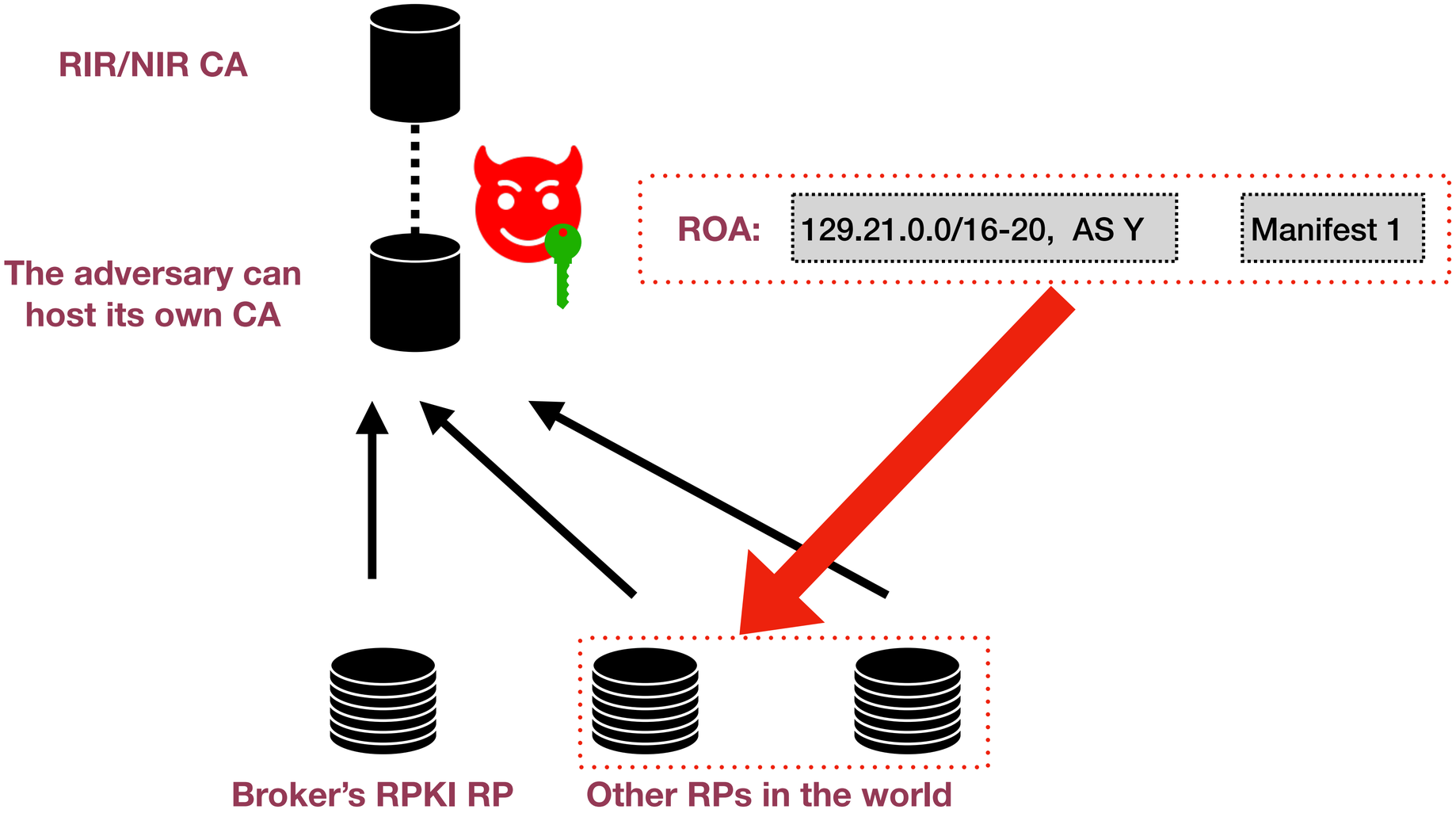

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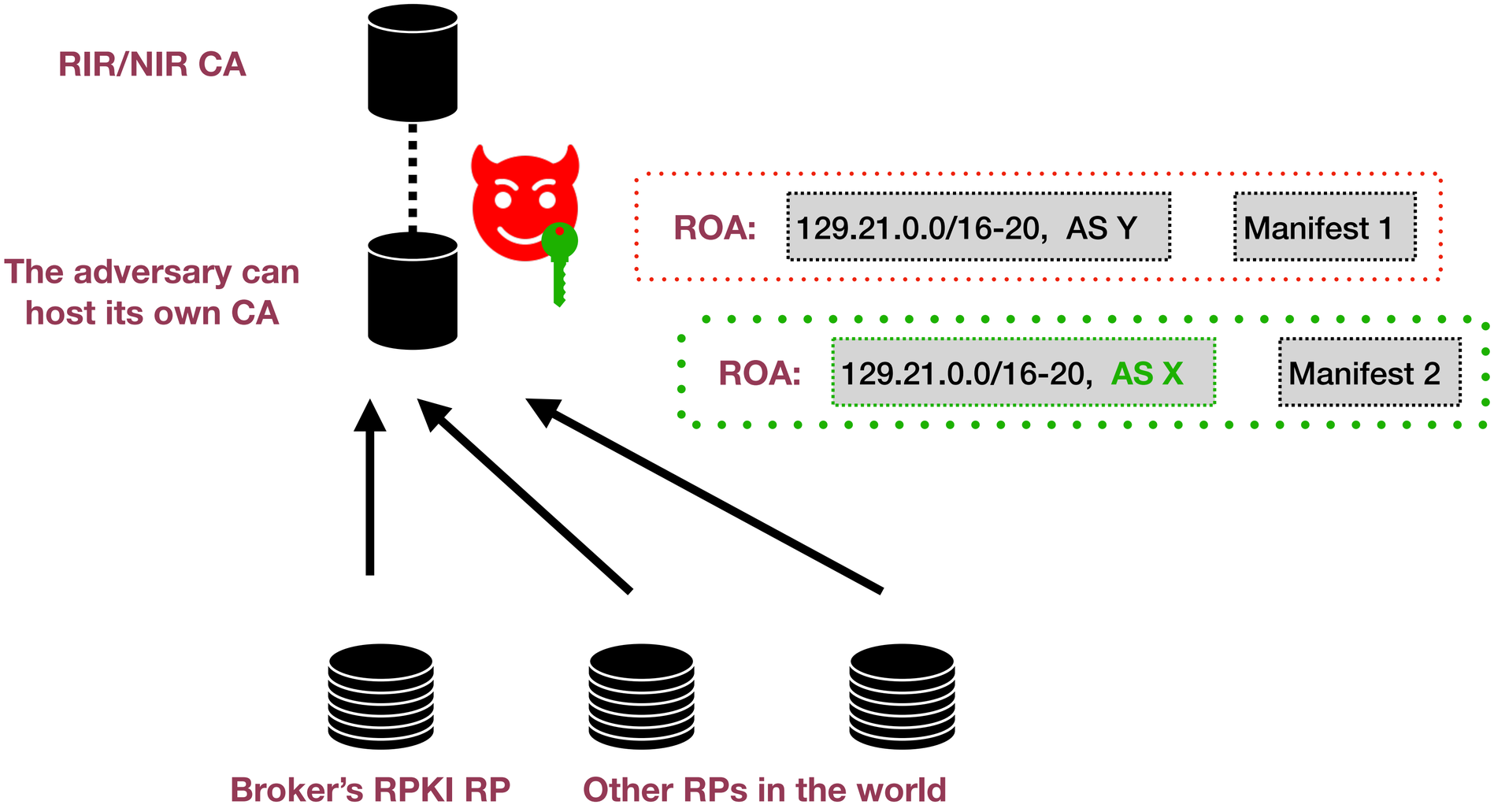

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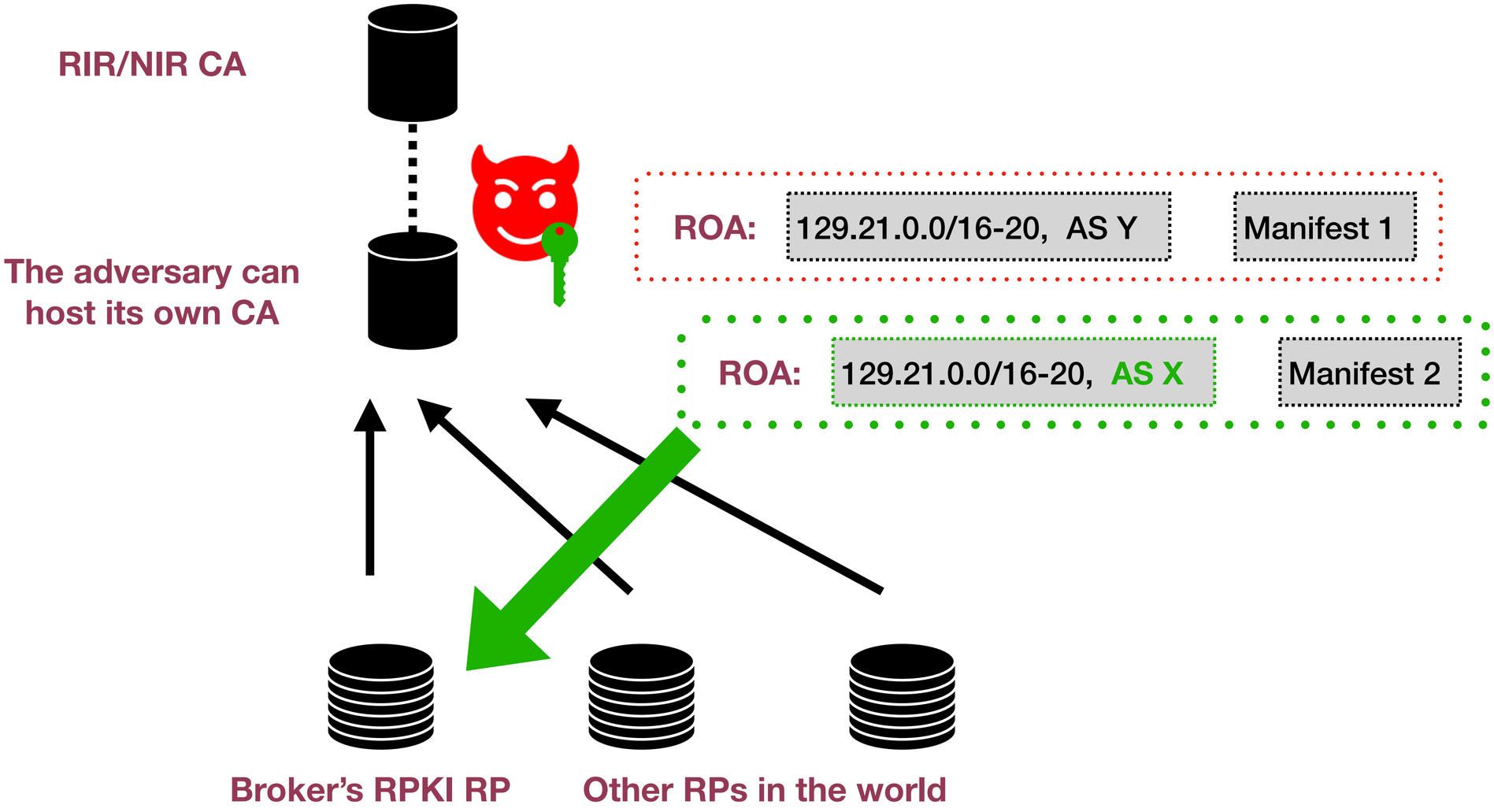

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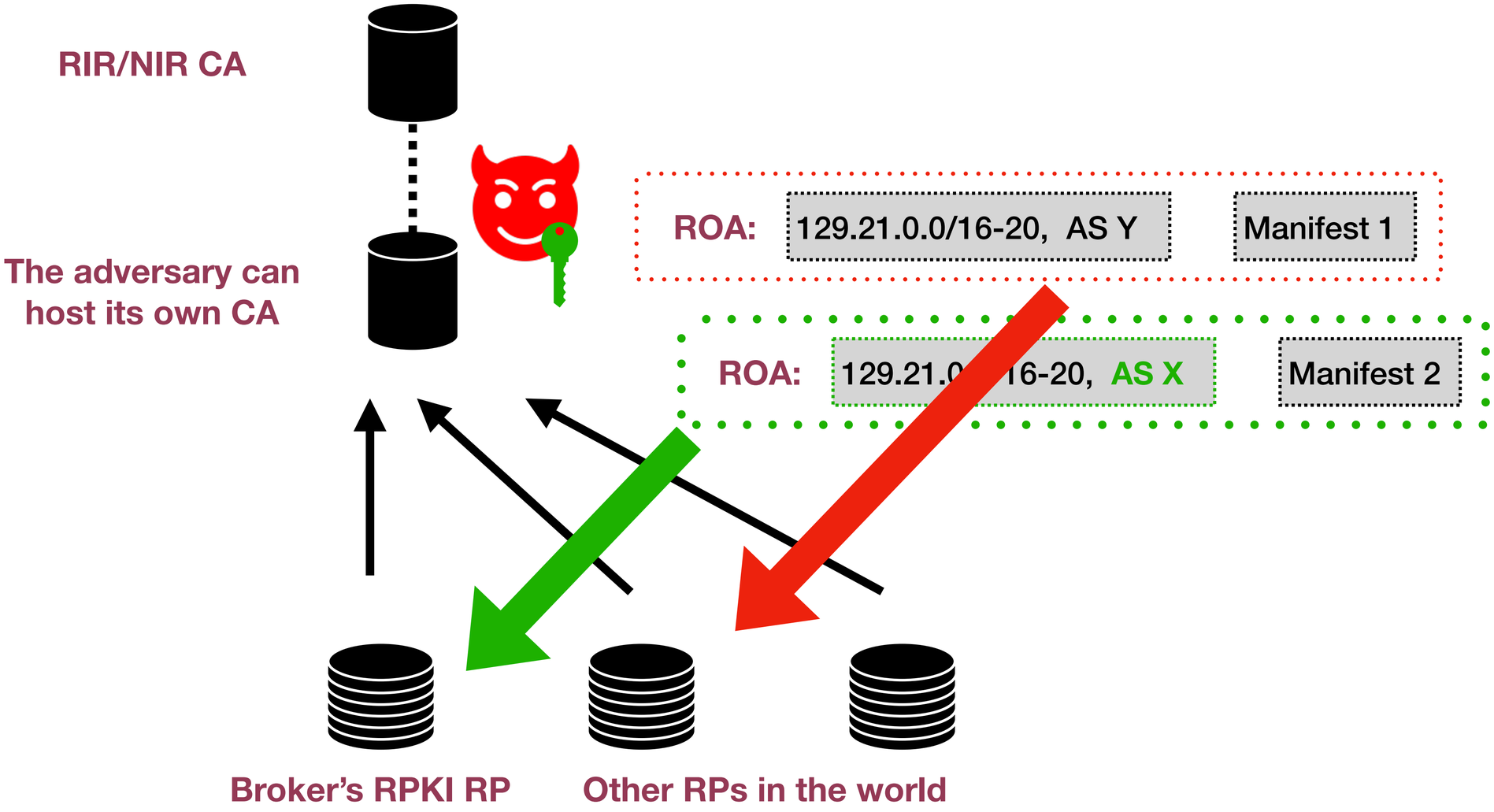
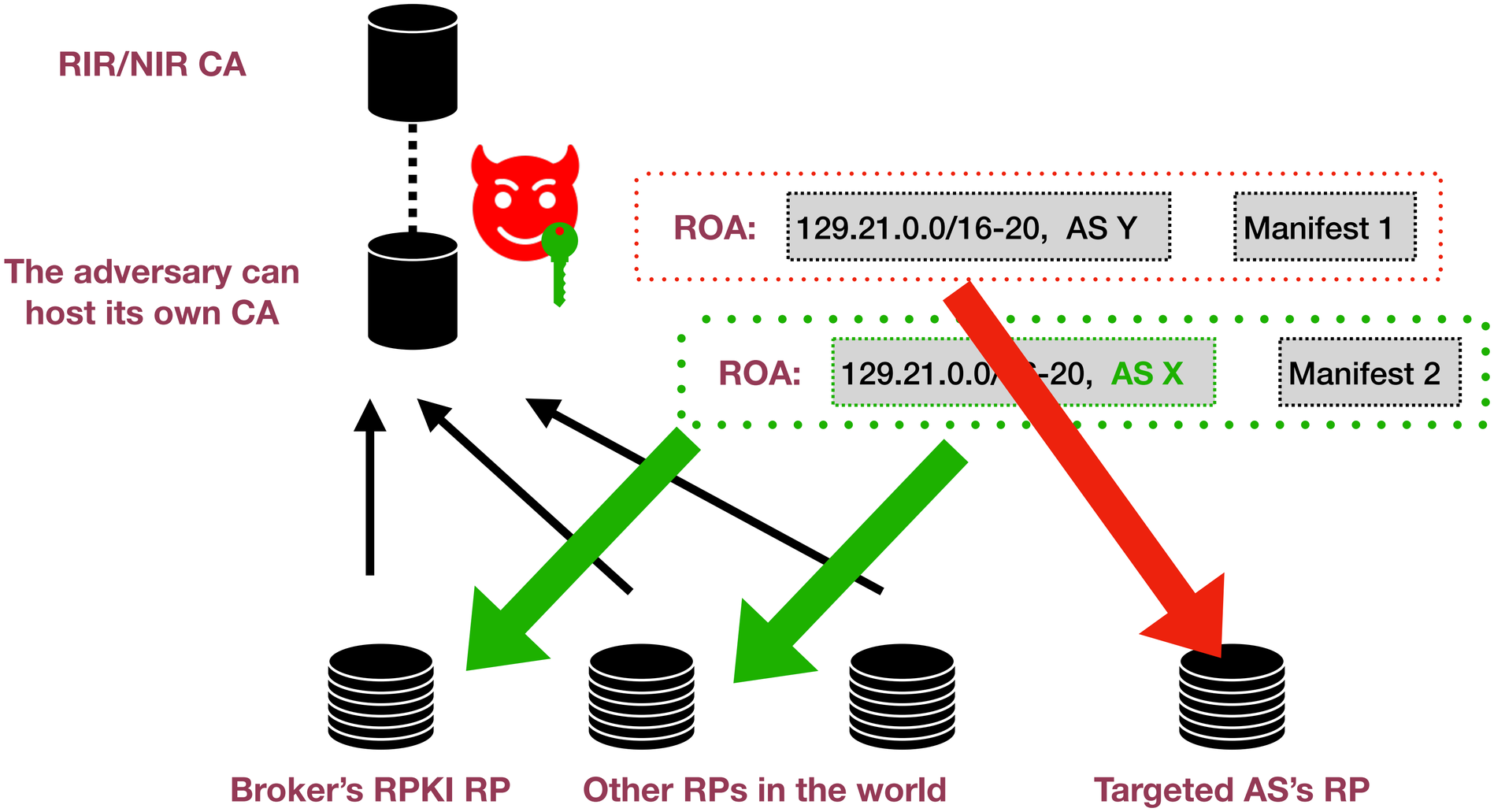


Targeted Specific Network(s) Only



Summary

- Why RPKI-Invalid happens: Misconfiguration is the major cause of RPKI-Invalid (**96.9%**), and the complexity of **transit and leasing business** making them hard to detect and mitigate.

- Impact of RPKI-Invalid: RPKI-Invalid could result in not only disconnection, but also path divergence, which is harder to detect but still **impacting security and performance**.

Summary

- Why RPKI-Invalid happens: Misconfiguration is the major cause of RPKI-Invalid (**96.9%**), and the complexity of **transit and leasing business** making them hard to detect and mitigate.

- Impact of RPKI-Invalid: RPKI-Invalid could result in not only disconnection, but also path divergence, which is harder to detect but still **impacting security and performance**.

- We need better management pipeline, better registration record delegation, and monitoring/automated tools to help ASes prevent misconfigurations

Thanks for Listening!

[NDSS'26] Demystifying RPKI-Invalid Prefixes: Hidden Causes and Security Risks
[Oakland'26] The Threat Landscape of IP Leasing in the RPKI Era

Weitong Li[1], Yongzhe Xu[1],Tao Wan[2], and Tijay Chung[1]

[1]Virginia Tech, [2]CableLabs